What Can Activation Patching Actually Establish? Reachable-State Geometry and the Limits of Mechanistic Evidence

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Abstract

Activation patching can produce strong, stable effects that still cannot tell a hypothesised mechanism from its nearby alternatives. We make this limit computable. Relative to a declared class of mechanisms, an intervention family has a *blind space*: the differences between hypotheses that vanish on every internal state the family reaches. The analyst chooses the intervention but the network produces the state, so the blind space can be computed without any readout. On analysed two-site operator slices, some hypothesis differences are exactly blind. The operator on the edge fixes the polynomial degree at which this begins: 2 for affine slices, 5 for the stated normalisation slices, and no finite degree for the analysed softmax and gated slices. In the tested charts and polynomial classes of trained transformers, the recompute single-site and joint designs have full rank, so the question becomes how weakly some directions are constrained. We re-analyse previously inspected captures using a hypothesis difference named from training intervention states without readouts. On held-out states, its mean square (energy) is at least $10\times$ smaller under single-site than under joint patching in $171/185$ captures across 10 model–task cells. The generator of the classical one-at-a-time design meets the same threshold in $144/185$. We also identify two pitfalls in comparing intervention families, and withdraw the claims they produced. On a published indirect-object-identification (IOI) edge, single-site evidence alone does not determine whether a head's measured effect carries over to other upstream states. Executing the missing four-point interventions, with the downstream head's output forced, measures this transport deviation directly. On the tested grids, its root mean square as a percentage of each capture's downstream-only readout range has median 0.427% in the exploratory analysis on three base prompts. In a preregistered replication on 24 fresh prompts from 8 templates, template medians met both registered smallness criteria. That replication could not compare published with control heads, because most control contrasts were not resolved above float32 precision, so we report no published/control ratio and make no above-control claim.

1 Introduction

Activation patching tests a mechanistic hypothesis by replacing an internal activation, letting the network recompute, and reading the change in behaviour as evidence that "head X contributes $+0.5$" or "A feeds B". Before reading the change that way, we can ask which alternative mechanisms the experiment could have ruled out. Recent work argues that interpretability claims should state their

An earlier version of this paper was accepted as a poster at the NeurIPS 2026 Workshop TAE (Trust-AI-Eval) and is public on OpenReview [Kadu et al., 2026]. This version corrects errors found after acceptance, which Appendix A records entry by entry. It also adds post-review analyses of the IOI result (§7) and a preregistered replication (Appendix O). Appendix N describes the released code, data and claim ledger.

Preprint.

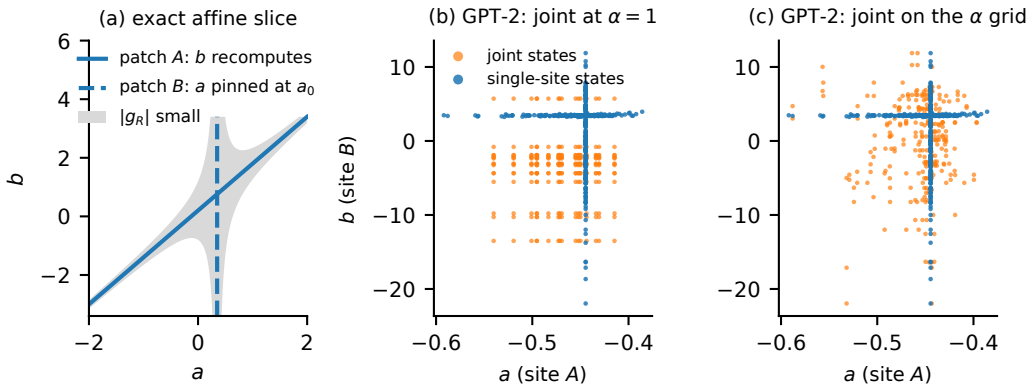


Figure 1: Why single-site patching can leave hypothesis differences weakly constrained; a and b summarise the two sites' outputs. *(a)* On an exact affine slice, single-site states lie on the blue recompute and pinned arms, where the hypothesis difference g_R vanishes. The grey region, where $|g_R|$ is small, contains both arms. *(b)* One GPT-2 small IOI capture (layer pair $(5, 10)$, seed 0, unembedding chart before whitening): single-site states (blue) form thin arms, while joint states (orange) at interpolation coefficient $\alpha = 1$ spread across a region. *(c)* Matching the joint family's α grid to the single-site grid broadens its range in this capture. The trained-model arms illustrate weak identification, not exact vanishing.

intervention class and identification assumptions [Joshi et al., 2026, Lin and Liu, 2026]. We make the intervention side of that question checkable from the states a model actually reaches.

Consider an edge $A \to B$, with one-dimensional summaries a and b (a *chart*) of the sites' outputs. A mechanism hypothesis predicts the readout as a function of these coordinates. Patching A lets B recompute, here along a line $b = ma + c$. Patching B leaves A at its clean value a_0. Single-site patching therefore visits two arms in the (a, b) plane. Two mechanisms that differ by $g_R(a, b) = (a - a_0)(b - ma - c)$ agree on both arms, because one factor vanishes on each, so no amount of single-site data can separate them. Joint patching sets both outputs and can reach states off the arms, where $g_R \neq 0$. For a non-flat edge, clamping can too: it holds B at its clean output while A is patched. Figure 1 shows the same picture in one GPT-2 small capture. There the recompute arm is a thin band while the pinned arm remains a line. This illustrates weak identification in a trained model, not exact blindness (§4).

The underlying mathematics is classical. For polynomial classes, blind spaces describe classical aliasing [Pistone and Wynn, 1996]. We compare intervention families by which hypothesis differences they resolve, using the support preorder of information matrices [Pukelsheim, 2006]. Recompute versus clamp parallels total versus controlled direct effects [Pearl, 2001]. Our contribution is to treat the network-produced internal states as the design points. In classical design the analyst writes them down. Under patch-and-recompute the analyst chooses the intervention and the *network* produces the internal state, so the aliasing structure is a property of the trained model that can be measured without observing any readout.

Section 3 establishes what those states can separate on idealised slices. Sections 4 to 5 study weak identification in trained models, and Section 6 examines misleading comparisons between designs. Section 7 asks what single-site evidence establishes about an effect at other upstream states. These questions require three distinct kinds of evidence.

Three kinds of claim

Exact slices (§3). On the analysed two-site operator slices, in the stated polynomial coordinates, the operator on the edge fixes the first degree with a nonzero blind difference. It is 2 for affine slices and 5 for the stated normalisation slices; the stated transcendental slices have no finite onset.

Trained models (§§4–6). In the tested charts and polynomial classes, the recompute single-site and joint designs have full rank. Some directions are nevertheless much more weakly constrained by single-site evidence than by joint evidence. This is weak identification, not exact non-identifiability.

Transport (§7). Without a model, single-site evidence does not determine whether a component's measured effect carries over to other states of an upstream component. Four-point interventions with the downstream output forced measure that transport deviation directly on the tested states.

We make four contributions. First, we formalise the blind space of an intervention family relative to a declared mechanism class, and the preorder it induces on families. We also separate what a single-site patch measures from the claim that this effect transports (§2).

Second, on the analysed two-site slices in the stated coordinates, we prove that the operator fixes the onset of exact blindness: $d^* = 2$ for affine slices, 5 for the stated normalisation slices, and $d^* = \infty$ for the stated transcendental slices. For affine single-site designs, the unresolved fraction of a polynomial class of fixed degree $d \geq 2$ also grows with the number of sites (§3).

Third, a direction named from training intervention states without readouts has at least $10\times$ less held-out energy under single-site than under joint patching in $171/185$ trained-model captures (§5). This re-analysis uses previously inspected captures; much of the weakness is already present in the classical one-at-a-time design. In our empirical re-analyses, support mismatch and chart selection each invalidated one of our comparisons of intervention families (§6).

Fourth, on a published IOI edge, single-site evidence cannot determine whether a head's effect transports (§7). For the exploratory analysis on three base prompts, with the downstream output forced, a cubic working model predicts the executed deviation worse than predicting zero. The median RMS deviation over captures is small on the tested grids. Template medians in a preregistered replication met both registered smallness criteria, but the published/control comparison remained unresolved, so the section remains exploratory. Table 1 maps the claims to their status and keeps the failed and withdrawn claims visible.

2 What a patching experiment observes

The opening example illustrates the principle: an intervention family sees a mechanism only through the states it reaches. Fix a model and k internal sites. A chart summarises the sites' outputs as a state $z \in \mathcal{Z}$, and a scalar readout y records behaviour (a logit difference throughout). An *intervention family* Q produces chart states: its reachable set $R_Q = \{z(q) : q \in Q\}$. A *mechanism hypothesis* is a function $f : \mathcal{Z} \to \mathbb{R}$ that predicts the readout. The *mechanism class* $\mathcal{F}$ is the set of alternatives the analyst agrees to consider. Here we take it to be a linear space of such functions.

Without a declared $\mathcal{F}$, "which mechanisms are distinguishable" has no answer, because any finite body of evidence is consistent with infinitely many functions. The blind space collects the hypothesis differences that vanish on every reachable state, and one family resolves at least as much as another when its blind space is contained in the other's.

Definition 1 (Blind space; resolving preorder)**.** The evidence map $\mathrm{Res}_Q : \mathcal{F} \to \mathbb{R}^{R_Q}$, $f \mapsto f|_{R_Q}$, restricts each hypothesis to the reachable set; its codomain is the space of real-valued functions on that set. Define

$$N(Q, \mathcal{F}) = \ker\big(\mathrm{Res}_Q|_{\mathcal{F}}\big), \qquad Q_1 \succeq_{\mathcal{F}} Q_2 : \Longleftrightarrow\ N(Q_1, \mathcal{F}) \subseteq N(Q_2, \mathcal{F}).$$

Two hypotheses are indistinguishable under Q iff their difference lies in $N(Q, \mathcal{F})$. Families are *incomparable* when neither relation holds.

Adding evidence can only shrink the blind space, since $N(Q_1 \cup Q_2, \mathcal{F}) = N(Q_1, \mathcal{F}) \cap N(Q_2, \mathcal{F})$.

Three kinds of intervention matter here. Our interventions replace a site's complete output. For a chosen donor prompt, the coefficient α moves that output a fraction of the way from its clean value to

Table 1: Main claims and all failed or withdrawn claims, with their status. *Proved*: a theorem on the analysed slices. *Exact*: a model-free consequence of which states were executed. *Empirical*: a re-analysis of captures inspected before these analyses were fixed. *Exploratory*: an analysis chosen after the workshop review. *Replicated*: a registered criterion met in the preregistered replication. *Floor-limited*: the registered comparison is unresolved because control contrasts are not resolved above numerical precision. *Failed*: a planned hypothesis whose criterion was not met. *Withdrawn*: retracted, either already in the workshop version (marked †) or here.

claim	status	where
On analysed slices, the operator on the edge fixes the onset d^* $(2, 5, \infty)$	proved	§3, App. B
For affine single-site designs at fixed $d \geq 2$, the unresolved fraction grows with the number of sites	proved	§3, App. B
A readout-free named direction is S-weak (weakly constrained by single-site relative to joint evidence) in $171/185$ captures; the one-at-a-time generator meets the same threshold in $144/185$	empirical	§5
Support mismatch inflates reverse-escape counts ($105/185$ before grid matching; $9/185$ after); optimising a chart on the scored states creates a selection effect	empirical	§6
Single-site evidence alone does not determine the IOI transport deviation	exact	§7
Under the cubic working model the IOI transport deviation is weakly constrained and poorly predicted	exploratory	§7
The RMS of the executed IOI transport deviation is small on the tested grids (median over captures on three base prompts; template medians on 24 fresh prompts from 8 templates)	exploratory; registered smallness criteria replicated ($8/8$, $8/8$ templates)	§7, App. O
Whether the IOI transport deviation exceeds the level of heads with no claimed role	floor-limited	§7, App. O
Mutual non-domination of single-site and joint evidence; empirical mutual non-domination of recompute and clamped single-site evidence in a chart optimised on the scored states	withdrawn†	§6, App. E, F
Specificity of single-site detection of non-additivity and its residual-bootstrap calibration; a reported number compatible with a much larger contribution; a representation no-separation claim; residual decay as a diagnostic of transcendence; auxiliary comparisons, angle stability, a Taylor check and numerical Davis–Kahan bounds	withdrawn	§7, App. A
Reciprocal cross-family degradation of a frozen circuit; representation-specific weak subspaces	failed	App. G

its value on the donor prompt: $\alpha = 0$ is clean, $\alpha = 1$ is the donor, and values outside $[0, 1]$ extrapolate. Under *patch-and-recompute*, patching A replaces its output and lets every downstream site recompute. On an analysed two-site slice, suppose B's recomputed chart coordinate is a function $b = e(a)$ of A's coordinate (the realised $A \to B$ map), and let a_0, b_0 be the clean coordinates. Single-site patching then has two arms. Patching A reaches $(a, e(a))$, and patching B with A clean reaches (a_0, b), so $R_S = \{(a, e(a))\} \cup \{(a_0, b)\}$. The graph relation is an idealisation of the analysed slice; in trained models, Sections 4 to 6 use the recorded intervention states directly. Under *clamping*, B is held at its clean output while A is patched. Together with patching B while A stays clean, this reaches $\{(a, b_0)\} \cup \{(a_0, b)\}$. *Joint* patching sets both outputs.

Every reported effect is conditional on its prompts, sites, replacement outputs and execution semantics. We therefore do not treat it as a context-independent contribution of a component, and §7 tests one such transport claim directly.

3 Exact geometry on operator slices

We first analyse the single-site families just defined on a two-site slice whose reachable set can be written down. Proofs are in Appendix B. Here the mechanism class is the real polynomials in (a, b) of total degree at most d, $\mathcal{F} = \mathcal{P}_{\leq d}$. The blind space is $N(Q, \mathcal{P}_{\leq d}) = I(R_Q) \cap \mathcal{P}_{\leq d}$, where the *vanishing ideal* $I(R_Q)$ collects all polynomials that vanish on the reachable set. We write $d^*(Q) = \min\{d : N(Q, \mathcal{P}_{\leq d}) \neq 0\}$ for the *onset* of exact blindness.

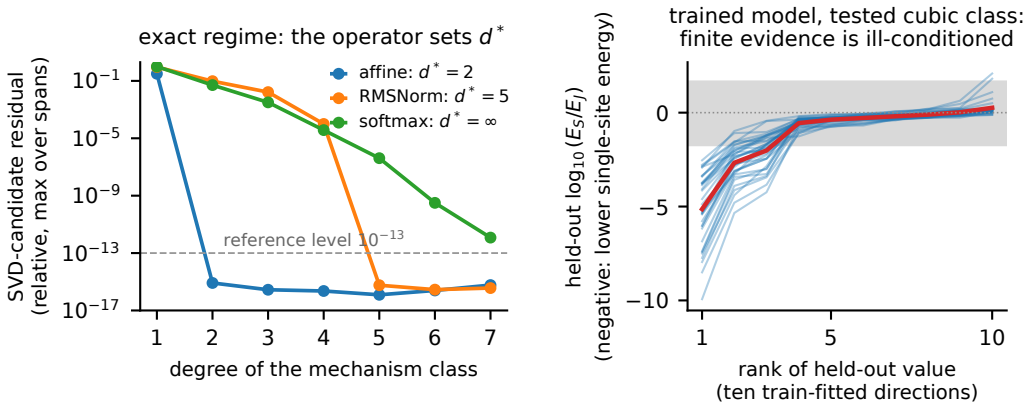


Figure 2: Exact operator slices and a trained model. *Left:* on each operator slice, singular value decomposition (SVD) fits a candidate polynomial relation separately on three sampling spans ($[-2, 2]$, $[-8, 8]$, $[-32, 32]$). The plot shows the maximum of their relative residuals. The affine and RMSNorm residuals reach numerical precision at their proved onsets. The softmax residual decreases over the tested degrees, but numerical residuals alone cannot distinguish exact relations from accurate approximations (Remark 4). The dashed line is a reference level of 10^{-13}. *Right:* GPT-2 small IOI captures, with the joint family on the same α grid as the single-site family (§4). Both tested cubic designs are full rank, but some train-fitted directions have much lower held-out energy under single-site than under joint interventions. The vertical axis is $\log_{10}(E_S/E_J)$, the logarithm of the inverse of the ratio r defined in §4, so negative values mean lower energy under single-site interventions. Each thin line shows one capture's values in increasing held-out order, and the thick line averages these ordered profiles; the grey band is the same-family envelope of §4, shown for display.

For the recompute single-site family on an affine edge $e(a) = ma+c$, with $b_0 = ma_0+c$, every blind polynomial is a multiple of the single *generator* $g_R = (a-a_0)(b-ma-c)$, so exact blindness begins at degree 2. The clamped family has the degree-2 generator $g_C = (a - a_0)(b - b_0)$. Appendix B shows that the two designs can have the same Hilbert profile (the number of independent constraints the evidence imposes) while missing different hypothesis directions (Proposition 5), so counts alone do not rank them. On this slice the generators differ by $g_C - g_R = m(a - a_0)^2$.

More generally, suppose the recompute arm lies on an irreducible curve $P = 0$. Single-site patching then cannot distinguish hypotheses that differ by a multiple of $(a - a_0)P$, so exact blindness starts at degree $1 + \deg P$. The theorem states the conditions.

Theorem 2 (Operator-slice theorem)**.** *On a two-site slice, let the pinned branch be $A_0 = \{(a_0, b) : b \in B_0\}$ with B_0 infinite. Let the recompute branch $\Gamma = \{(a, e(a))\}$ be an infinite subset of the zero set $V(P)$. Assume that $P \in \mathbb{R}[a, b]$ is irreducible (not a product of two nonconstant polynomials) and not an associate (a nonzero constant multiple) of $a - a_0$. Then $I(A_0 \cup \Gamma) = \big((a - a_0)P\big)$ and $d^* = 1 + \deg P$.*

Affine edges have $\deg P = 1$, so $d^* = 2$. For the normalisation slices $e(a) = \beta+\omega a/\sqrt{\kappa a^2 + \epsilon}$ with $\omega \neq 0$, $\kappa > 0$ and $\epsilon > 0$, clearing the radical gives an irreducible P of degree 4, so $d^* = 5$ on these RMSNorm and LayerNorm slices: normalisation *raises* the onset. This is an exact statement about those slices, not about trained normalisation edges in general. If no nonzero polynomial vanishes on the graph of e, then $I(\Gamma) = (0)$, the blind space is zero at every finite degree, and $d^* = \infty$; a separate lemma proves this for two-key softmax and sigmoid-gated linear-attention slices with nonzero parameters.

With k sites patched one at a time, the reachable set is a union of k curves. On arm i the patched coordinate is swept, upstream coordinates stay clean and downstream ones are recomputed from it. Assume that every coordinate on each arm is a polynomial in that arm's swept coordinate.

Proposition 3 (Restriction-rank bound for polynomial single-site arms)**.** *Let the coordinates on arm i be polynomials $z_{ij}(t)$ of the swept coordinate t, with $z_{ii}(t) = t$, and let all arms pass through the*

clean configuration. Set $\eta_i = \max\big(\{1\} \cup \{\deg z_{ij} : z_{ij}$ *nonconstant*$\}\big)$. *Then* $\operatorname{rank}(\mathrm{Res}_Q|_{\mathcal{P}_{\le d}}) \le 1 + d\sum_i \eta_i$, *so* $\dim N(Q, \mathcal{P}_{\le d}) \ge \binom{d+k}{k} - \big(1 + d\sum_i \eta_i\big)$.

The degrees must include all downstream recomputation (Appendix B). For affine edges $\eta_i = 1$ and the bound $1 + kd$ is attained, while the polynomial class grows much faster with the number of sites. At degree 3, the unresolved fraction rises from 30.0% at two sites to 71.4% at five. A joint Cartesian product with at least four distinct values on each axis resolves all of it.

Remark 4 (Continuum, not grid; slice, not network)**.** Exact d^* statements concern the operator's continuum of reachable states: every finite grid admits vanishing high-degree polynomials, so numerical checks corroborate but never establish these algebraic claims. The statements concern analysed two-site slices in the chosen polynomial coordinates, *not* whether a full transformer has or lacks exact blind mechanisms.

In the trained-model charts and polynomial classes we evaluate, the recompute single-site and joint designs have no exact blind direction. The next section therefore asks how weakly they constrain different directions.

4 From exact blindness to weak identification

The exact results above concern operator slices; Sections 4 to 6 turn to trained models. For the recompute single-site and joint designs, in every capture, chart and class we evaluate, the exact blind space is trivial: both designs have full rank. The question is therefore graded: how much less does a hypothesis difference vary under one family than under another? These are statements about weak identification in the evaluated charts and classes, not about exact non-identifiability. The clamped comparison family of §6 is the one exception, because its pinned coordinate arms give it an exact polynomial kernel by construction (Appendix D).

We measure how visible a hypothesis difference is by its mean square on the states a family reaches. Let μ_Q be the measure that weights the states of Q, let φ be a feature map spanning $\mathcal{F}$, and let $G_Q = \mathbb{E}_{\mu_Q}[\varphi(z)\varphi(z)^\top]$ be the intervention Gram operator. A hypothesis difference $h = \theta^\top\varphi$ then has *energy* $E_Q(h) = \theta^\top G_Q \theta = \mathbb{E}_{\mu_Q}[h^2]$. Small energy means that the hypotheses differ little on those states. Under the regularity conditions that hold for the polynomial classes used here (continuous features, finite second moments, and $R_Q = \operatorname{supp}\mu_Q$), the kernel of G_Q is exactly the blind space. Energy therefore gives a graded analogue of exact blindness.

For a fixed hypothesis difference, or direction, h we report the *held-out* J/S *energy ratio* $r(h) = E_J^{\mathrm{te}}(h)/E_S^{\mathrm{te}}(h)$, where S and J are the single-site and joint families. A large ratio means that single-site evidence constrains h much less than joint evidence does, and we call such a direction *S-weak*. "Held-out" means that the direction is fitted or named on training rows and scored on the complementary rows: the rows of each design are split in half, and directions are frozen before scoring. For the named direction of §5, the edge fit, whitening and feature scaling use training rows only. The archived optimised profiles instead used all analysed rows for whitening and scaling. For the fixed polynomial basis and covariant eigensolve, Appendix D shows that this is only a change of feature basis in exact arithmetic and leaves the held-out profiles unchanged. To find directions with contrasting energies, we form the Gram operators on training rows and solve the regularised generalised eigenproblem

$$\big(G_S + \lambda G_{\mathrm{ref}}\big)w = \nu\,\big(G_J + \lambda G_{\mathrm{ref}}\big)w, \qquad G_{\mathrm{ref}} = \tfrac{1}{2}(G_S + G_J), \quad \lambda = 10^{-6},$$

which regularises both sides by the same covariant reference operator, so that a change of feature basis leaves the spectrum invariant (Appendices D and K). The held-out profile orders the log single-site-to-joint energy ratios of all frozen training eigen-directions on test rows. Its low end is S-weak; its reverse end is joint-weak.

We analyse 185 float32 captures across 10 model–task cells; a capture is one run for a pair of sites and a seed. Captures share prompts and tasks, so counts are consistency checks and we make no population claim. The degree-3 polynomial class, the intervention grids and the chart are described in Appendix D. A readout screen excludes captures with $\mathrm{sd}(Y_{\mathrm{joint}}) < 0.15$; it is the only use of readouts in Sections 4 to 6, and it makes the analysed sample readout-conditioned.

Optimising a direction can produce extreme ratios even within one family. We therefore rerun the train–freeze–score procedure on two splits of the single-site family and use their held-out profiles

as a *same-family envelope*, the null band for this comparison. Because these splits are smaller and captures share prompts, escape counts are descriptive and any p-values are nominal; Appendix D gives the exact construction.

5 A weak direction predicted from intervention geometry

Is single-site weakness only an artefact of searching for a weak direction? To find out, we score the one direction the theory names without any search. On an affine slice the single-site blind space is generated by $g_R = (a - a_0)(b - ma - c)$. We estimate it from intervention states alone: from the training rows, a_0 comes from the downstream-patching arm and (m, c) from a least-squares fit of b on a along the upstream arm. No readout enters, and the direction is not chosen to separate the families. This is a re-analysis of captures we had inspected earlier with an optimised procedure; we ran no fresh validation of the named direction.

The held-out J/S energy ratio of this direction exceeds $10\times$ in $171/185$ captures across 10 cells. Appendix H gives the per-cell medians (Table 5) and three further checks (donor indices held out, summaries by prompt group, random directions), which give the same qualitative picture. A classical one-at-a-time baseline explains much of this. The clamped generator $g_C = (a - a_0)(b - b_0)$ is the blind generator of the one-at-a-time design, and it vanishes on both of that design's arms. On an exact affine slice it equals $m(a - a_0)^2$ on the recompute arm. It is small when the edge is flat and the scatter low, and most edges in this chart are flat (Appendix E gives the finite-capture expression). Scored on the same recompute single-site and joint states, g_C also clears $10\times$ in $144/185$ captures. The recompute generator exceeds this baseline count by a net 27 captures.

Relative to the one-at-a-time baseline, the recompute generator becomes more distinctive as the fitted edge gets steeper: $r(g_R)/r(g_C)$ increases with the absolute whitened edge slope (Spearman 0.587). Its absolute ratio $r(g_R)$ nevertheless falls as the edge strengthens (Spearman -0.869 over 185 captures), where edge strength s is $\mathrm{sd}(b \mid \text{arm } A)/\mathrm{sd}(b \mid \text{arm } B)$ on training single-site rows. The largest ratios therefore come from captures where A barely drives B. Our attempt to remove this dependence by choosing the chart failed (§6); Appendix E gives the remaining edge diagnostics.

What does the construction guarantee? On an analysed algebraic slice, the two single-site arms satisfy equations whose product vanishes on the whole reachable set, which is why a candidate can be named. Whether the fitted candidate is weak in a finite trained-model sample, and by how much, is an empirical question. The normalisation onset 5 applies only to the analysed normalisation slice, not to a trained edge. Appendix J checks the estimator on synthetic data and Appendix I reports a three-site experiment.

In these captures, single-site evidence has a weakly constrained direction that geometry can name from one sweep of intervention states. That does not show that the single-site family is worse than every other family, and such comparisons are easy to get wrong, as the next section shows.

6 Two traps when comparing intervention families

The comparison above raises a further question: could intervention families look differently informative for reasons unrelated to the mechanism of interest? We encountered two such artefacts in our own comparisons. The corrected analyses follow, and the claims they overturned are withdrawn (Table 1).

Unmatched supports. Our single-site family sweeps $\alpha \in [-0.5, 1.5]$, extending half the clean-to-donor distance beyond each endpoint, while the joint family used raw donors ($\alpha = 1$). A polynomial that is large only in the extrapolated region then looks "visible to S, weak under J" for reasons unrelated to any mechanism. The same-family null cannot detect this, because both halves of S share the same support. This mismatch underlay our withdrawn claim that single-site and joint evidence are *mutually non-dominating*, that is, that a reverse direction also exists, weakly constrained by joint evidence and well constrained by single-site evidence. We therefore recollected the joint family on the same α grid, with the same prompts, donors and weights. Reverse escapes above the same-family envelope fell from $105/185$ captures to $9/185$, and restricting both families to $\alpha \in [0, 1]$ leaves $22/185$. The few remaining reverse escapes are descriptively associated with stronger edges (Appendix E). The matched joint family is a sparse design on the same grid, not a product grid, so

exact support matching is not established; the comparison uses paired float32 captures. The lesson: before comparing two intervention families, check that their reachable sets span the same region. Interpolation grids make it easy to compare an extrapolating family with an interpolating one and read the difference as mechanism.

Selecting a chart on the evaluation states. To make the recompute and clamped single-site designs differ, the edge must be non-flat, and in the unembedding chart it is nearly flat (median absolute whitened slope 0.0359). We therefore selected the downstream coordinate by maximising a readout-free, regularised variance ratio between the two arms, and compared recompute with clamped single-site evidence in that chart. The recompute-weak end of the optimised profile then escaped the same-family envelope in $103/170$ captures. With a chart fitted on a different seed of the same layer pair it escaped in $26/120$. The two analyses cover different cohorts (Appendix F), so this is not a paired effect estimate. The selection rule helps explain the failure. For GPT-2 its denominator was estimated from at most 16 donor directions in a 768-dimensional activation space. In the GPT-2 captures for which this was logged, the selected direction had numerically negligible projection onto the span of those donor directions. A different seed is not a disjoint-content validation either: some evaluated captures share the chart source's base prompt, and donors come from a common pool. We withdraw the claim of non-domination in a chart optimised on the scored states. The lesson: optimising a chart on the states where it is scored is a selection effect, even when no readout is consulted. Fit the chart on different prompts and donors from those on which it is scored.

These traps show why an intervention comparison must specify the states it reaches. The next section asks what those states can establish about a single measured effect, the kind a circuit analysis reports.

7 Does a measured effect carry over? A published IOI edge

Does a Name Mover head's measured single-site effect carry over to other states of an upstream S-Inhibition head? We ask this on the S-Inhibition $\rightarrow$ Name Mover edge of the IOI circuit of Wang et al. [2023] in GPT-2 small. Interpreting a single-site effect as a contribution that persists across upstream states assumes that it does. This is the practical form of the blind-space question, because a single-site run never visits the states where the effect would have to hold. Because our test forces the Name Mover's output, it asks only how the readout combines the two heads' outputs on the executed states. It does not test the query-mediated pathway by which S-Inhibition changes what the Name Mover writes, and it does not contradict the mechanism Wang et al. [2023] identified by path patching.

The transport contrast

The question is simple: does B have the same measured effect after A changes? Write u and v for replacement outputs of A and B, and 0 for each site's clean output (so $u = 0$ corresponds to $a = a_0$). $Y(u, v)$ is the readout with A's output set to u and B's output forced to v; $Y(u, 0)$ keeps B at its clean output. Patching B alone measures its effect *in the context* where A is clean,

$$\Delta_B(v) = Y(0, v) - Y(0, 0).$$

If this effect transports to the upstream state u, the same $\Delta_B(v)$ should apply once A is changed to u. Exactly, and without any model,

$$Y(u, v) - Y(u, 0) = \Delta_B(v) + T_C(u, v), \qquad T_C(u, v) = Y(u, v) - Y(u, 0) - Y(0, v) + Y(0, 0).$$

The four-point (difference-in-differences) *transport contrast* T_C uses the clamped corner $Y(u, 0)$, and it vanishes at every tested (u, v) exactly when B's measured effect carries over to every tested state of A. Patching A alone observes $Y(u, v_{\mathrm{rec}}(u))$, where $v_{\mathrm{rec}}(u)$ is the output B recomputes to; it generally observes neither the joint corner $Y(u, v)$ nor the clamped corner $Y(u, 0)$. Single-site evidence therefore does not determine T_C without additional assumptions. Executing the two missing corners measures it directly and needs no chart.

Design. We pair four S-Inhibition heads with three Name Movers (twelve published head pairs). Controls pair four early heads that have no claimed IOI role with the same Name Movers. A *head pair* with a prompt is a capture; on three base prompts, 31 of the 36 published captures and 28 of the

36 control captures pass the screen. Each capture executes the single-site arms, the clamped arm and 256 joint states; Appendix L lists the head identities.

The executed contrast is small on the tested grids. This analysis was chosen after seeing the data, so we call it exploratory. For each capture on the three base prompts we compute the root mean square (RMS) of T_C over the tested joint states. The median RMS over captures is 0.00709 logits. Dividing each capture's RMS by its *B-arm swing* gives a median percentage of 0.427%. The swing is the readout range over the single-site B arm (32 donors $\times$ 7 coefficients; median 1.65 logits, never below 0.67). The largest single $|T_C|$ in a capture is 0.30 logits at most, so smallness describes the RMS over states, not every state. These captures do not measure the numerical floor; Appendix L gives the spread and a second normalisation.

A preregistered replication meets both registered smallness criteria. Before collecting fresh data, we preregistered a replication (Appendix O). It used 24 new prompts, three from each of 8 IOI templates of Wang et al. [2023] that our earlier captures had not used, with the same head pairs, controls, design and analysis. For each prompt we take the median RMS of T_C over its screened published captures. For each template we then take the median of its three prompt medians (V_t), and likewise for the percentage of the swing (V_t^{rel}). V_t was below 0.10 logits in $8/8$ templates and V_t^{rel} below 1% in $8/8$, so both registered smallness criteria passed. Both margins were set after the exploratory analysis, and the relative one is the informative criterion. Registered smallness concerns template medians, not every capture; individual captures can be larger (Appendix O).

The control comparison is unresolved. The replication also measured the float32 floor by re-executing every corner family in a different batch composition. Published contrasts exceeded ten times their floor in $281/281$ captures, whereas control contrasts did so in only $111/268$. The registered comparison of published and control contrasts (H2) was therefore floor-limited under the registered rule. We report no published/control ratio and make no above-control claim, and under the registered decision rule this section remains exploratory, with its smallness replicated as registered.

What single-site evidence leaves unmeasured. Single-site interventions do not execute all the corners that define T_C, so without a model they cannot determine it. What can a model recover? The exploratory working model uses a fixed, weight-derived chart motivated by the published claim that S-Inhibition heads act on the Name Movers' *query*. The upstream coordinate a is the S-Inhibition head's output along the leading singular direction of $W_O^{\mathrm{SI}} W_Q^{\mathrm{NM}}$, the composition of its output weights with the Name Mover's query weights. The downstream coordinate b is the Name Mover's output along its direct effect on the logit difference. Neither direction uses intervention states or readouts. The working model is the cubic class in this chart, with additive terms in a or b alone and cross coefficients γ on ab, a^2b and ab^2. Under it $T_C(u, v) = D(u, v)^\top \gamma$, where D is the four-point contrast of the cross monomials, and the additive terms cancel. On the pinned arm every contrast vanishes, and on the recompute arm of an affine edge the cross terms reduce to functions of a alone, which the additive terms absorb. Single-site data therefore inform γ only through the scatter of b about the fitted edge. After residualising the cross terms against the additive terms, we compare the smallest eigenvalues of the per-observation design Grams in this shared chart. The joint eigenvalue is far larger than the single-site eigenvalue. Their joint/single-site ratio is the *design gap*; Appendix L reports its magnitude. It is a property of the design, not Fisher information about the network.

On the exploratory captures, the working model does not predict the executed contrast. Fitted on single-site states, the cubic predicts held-out joint readouts far worse than a constant (median R^2 -439), while an additive-only refit transfers (median R^2 0.70). Its median held-out RMS error in predicting T_C is 7.45 logits, against 0.010 for predicting zero. The bootstrap interval for the excess over predicting zero lies wholly above zero in $31/31$ single-site and $22/31$ joint fits. The model is misspecified on both supports; only the executed contrast is model-free. Table 6 in Appendix L gives the fit statistics.

For each joint state we also compute the largest $|T_C|$ inside the fit's nominal 95% Wald region for γ under the working model, and summarise these by their median within each capture. We call this a working-model identification width. Because the cubic is misspecified, it is neither an observed error nor a calibrated bound: the pointwise envelopes contain the executed $|T_C|$ at every joint state in

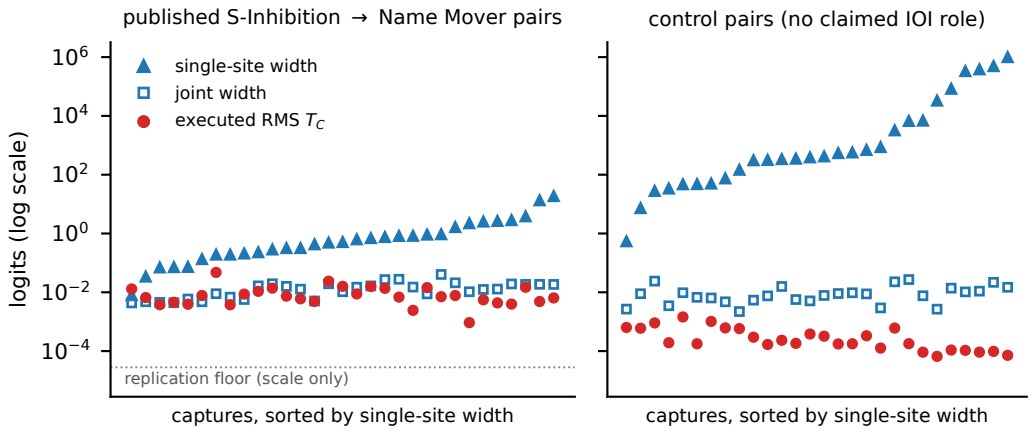


Figure 3: Working-model identification widths and executed transport contrasts, per capture (exploratory; three base prompts). Left: published S-Inhibition → Name Mover captures; right: control captures. Captures are sorted separately within each panel. Triangles and open squares show the single-site and joint working-model identification widths, each a median over joint states; circles show the RMS of the executed T_C over those states (logarithmic scale, logits). For the published heads, the median single-site width divided by the median executed RMS is 72.1: the executed RMS is much smaller than the single-site width. The widths are properties of the misspecified working model, neither observed errors nor calibrated bounds. The exploratory control captures did not measure their numerical floor. The dotted line is only a scale reference, the median floor measured in the replication for its published captures. The plot supports no above-control conclusion.

only $12/31$ (single-site) and $1/31$ (joint) captures. Against the B-arm swing, the median single-site working-model identification width is 31.1% and the joint one 0.764%.

Single-site rejections did not detect the executed deviation. The workshop version read the rejections of $\gamma = 0$ by single-site evidence as detection that separates published from control heads. That specificity depends on the covariance estimator, and we withdraw it; Appendix L gives the counts under alternative covariances. More fundamentally, because the cubic is misspecified on both supports, rejecting its additive restriction is not a measurement of the executed transport contrast. Figure 3 makes the mismatch visible: the single-site working-model width is far larger than the executed RMS deviation, while the joint width is closer in scale but poorly calibrated. Neither can be read as a resolved measurement of T_C.

Scope. Everything here concerns GPT-2 small, the published S-Inhibition → Name Mover head pairs, output-replacement interventions and the tested grid of states. Without executing the corners one cannot know whether a transport deviation is small or large. Measuring it requires the clamped and joint corner families in addition to the single-site arms. Appendix L also reports a second head group, previous-token and induction heads. Induction heads read previous-token information through keys, so a final-position query chart does not test that mechanism, and we do not treat the group as a second query-mediated circuit.

8 What an audit should record

The geometry and IOI analyses give a practical checklist for auditing new patching evidence, together with the limits that qualify our results.

- **Level.** The exact theorems apply only to the analysed operator slices and polynomial single-site arms, under their stated coordinates and assumptions (Remark 4, Proposition 3). The trained-model results concern weak identification in the evaluated charts and classes, not exact non-identifiability; the clamped family's polynomial kernel is structural.

- **Class and chart.** $N(Q, \mathcal{F})$ depends on $\mathcal{F}$ and the chart by construction, so the framework requires a declared protocol and gives no context-free result.
- **Scope and selection.** The trained-model geometry analyses use 185 two-site captures across 10 model–task cells. Captures share prompts and tasks, the joint design is sparse rather than an exact product-support match, and the readout screen conditions the analysed sample. Counts are therefore descriptive, and we make no population claim.
- **Named direction.** The analysis reuses previously inspected captures, and the one-at-a-time baseline explains most threshold exceedances.
- **IOI.** The study covers GPT-2 small and one published edge class. The executed contrast is exploratory on three base prompts. In the preregistered replication on 24 fresh prompts from 8 templates, H1 (absolute) and H1 (relative) passed in $8/8$ and $8/8$ templates. H2, the published/control comparison, was floor-limited, so we report no published/control ratio or above-control claim. Smallness refers to capture medians on the tested exploratory grids and to registered template medians on the replication grids. The cubic working model does not predict the executed contrast, and its identification widths are uncalibrated. T_C forces the downstream output and does not test the query-mediated pathway.
- **Inference.** p-values from the same-family comparison are nominal, and robust-covariance counts are sensitivity analyses. Two planned hypotheses failed (Appendix G).

An audit checklist for patching evidence

1. Declare the mechanism class and the chart before using readouts to tune either of them.
2. Record the intervention *states* as well as the readouts. For each declared mechanism class, inspect the family's Gram operator and its low-energy directions (Appendix D).
3. Match the regions that families reach before comparing them, and report any remaining mismatch. Fit any data-derived chart on prompts and donors distinct from those used for scoring.
4. Before interpreting model-based intervals, test the working model on held-out executed contrasts and check its calibration. Keep hypothesis rejections separate from effect estimates.
5. For a transport claim, declare the tested states, the aggregation unit, and absolute and relative margins. Execute the four-point contrast and measure its numerical floor by re-executing it under a different batch composition.
6. Record when the class, chart, direction and decision criteria were fixed relative to data inspection, and distinguish exploratory re-analysis from preregistered evaluation.
7. Account for unequal arm variances and donor dependence. Use inference units appropriate to the design (heads, prompts or templates) and state the dependence that remains.

An activation-patching experiment supports a mechanistic claim only relative to the alternatives its reachable states can separate. Within a declared mechanism class, the reachable states determine which alternatives are exactly unresolved, and their weighted geometry reveals directions that are only weakly constrained; no readout is needed for either. Transport is a separate question. To assess whether a per-component effect transports, execute the missing corners, because better single-site statistics cannot determine it without a model.

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A Correction log

The version accepted at the workshop [Kadu et al., 2026] contained errors that we found after acceptance. Each row below is one entry of a correction ledger (57 entries). For each entry the ledger records an independent verification check appropriate to the correction (recomputation, mathematical checking or inspection) and, where applicable, the source artifacts, scripts and cited files by hash and the command or inspection behind the correction; associated review files record the adversarial verdicts. This table is generated from that ledger. The last column states whether a scientific conclusion changed. All other numbers in this paper are the corrected ones.

id	item	accepted version	this version	cause	conclusion changed?
C01	Named-direction count and cell medians (Table 5)	$172/185$ captures above $10\times$; Qwen3-8B medians 86.6 (IOI) and 91.4 (induction)	$171/185$; 60.0 and 66.4; every cell median recomputed with the edge fitted on training rows	the edge (a0, m, c) was fitted on rows that included the held-out half (the whitening leak cancels from this ratio)	No (procedure corrected)
C02	Random-direction null (Table 5)	95th percentiles 3.99–14.0	cell medians of capture-specific 95th percentiles 7.02–11.4 on the same captures and split; the named direction beats its own capture's null in $180/185$	null range taken from different (unmatched, raw-donor) captures and an all-rows fit	Magnitude only
C03	Donor-level robustness	$168/185$ captures under a "stricter whole-donor split"	that test scored the optimised direction and counted escapes of the same-family envelope; with both donor indices held out, the named direction's held-out J/S energy ratio is above $10\times$ in $173/185$ captures; donor content is not held out (identical patch-A trajectories cross the partition in $72/185$ captures)	a different direction, partition, null and criterion were reported as the named direction's donor test	No (procedure corrected)
C04	Prompt-level robustness	$25/38$ prompt groups; bootstrap lower bound 0.856	row-split named direction: every capture above $10\times$ in $28/38$ (cell, prompt) groups; a bootstrap over groups gives 87.4% as the lower end of a 95% percentile interval for the pooled fraction of captures above $10\times$	prompt-level analysis scored the optimised eigenvector, not the named generator	No (procedure corrected)
C05	IOI envelope: the four-point contrast	single-site 42.9% vs joint 0.779% (ratio 55.1) "of the reported number"; exceeds the whole number in $11/31$; synthetic self-test coverage 97.7%	working-model Wald envelopes of 31.1% vs 0.764% of the B-arm swing (40.6); the single-site envelope exceeds the whole swing in $9/31$; envelopes contain the executed T_C at all states in $12/31$ and $1/31$ captures; synthetic self-test coverage 97.3%	the code evaluated two of the four terms of the stated contrast	Yes: interpretation
C06	Denominator of the IOI percentage	"of the reported number", read as $Y(a_0, b) - Y(a_0, b_0)$	the B-arm swing (median 1.65, minimum 0.67 logits)	the prose named a different denominator from the one the code used	Yes: interpretation

id	item	accepted version	this version	cause	conclusion changed?
C07	Size of the contribution a reported number is compatible with	a per-component number on these heads is compatible with a substantially larger contribution	withdrawn; post-review exploratory analysis on three base prompts: the executed T_C is small on the tested grid (median RMS 0.00709 logits, 0.427% of the B-arm swing), and the single-site fit predicts it worse than zero in $31/31$; the preregistered replication on 24 fresh prompts from 8 templates met the absolute and relative smallness criteria for template medians of prompt medians over screened published captures on its tested grids ($8/8$ and $8/8$ templates, respectively); the control family was floor-limited ($111/268$ control captures resolved, failing the registered resolution gate), so no above-control comparison is made	an upper envelope under a working model that does not transfer was read as a plausible contribution	Yes: claim withdrawn
C08	Specificity of single-site detection	published $26/31$ vs controls $0/28$ reject; "not calibration failure"	under alternative covariances published 26–29, controls 8–16, induction group 7–10; covariance-sensitivity calculations for a misspecified model, not a validated calibration	one pooled homoskedastic residual variance across arms with very different variances	Yes: claim withdrawn
C09	Residual-bootstrap calibration argument	null size 6.42% (single-site), 5.9% (joint)	residuals pooled across arms whose variance ratio is 0.522 (published) and 0.00231 (controls); argument withdrawn	the null resampled residuals across heteroskedastic arms	Yes: claim withdrawn
C10	Joint coefficient "consistent with zero"	$\lvert c\rvert/\mathrm{se}$ of 1.08 (IOI), 0.91 (induction), 0.979 (controls)	a ratio of norms, not a test; with the marginal-sum donor covariance $V_{\mathrm{up}} + V_{\mathrm{down}}$ (no intersection term or cross-role dependence; calibration unknown) the joint Wald test rejects in $1/31$ published and $1/28$ control configurations	a norm ratio was read as a t-statistic	Yes: interpretation
C11	Pseudo-replicated p-values	edge contrast $p = 2.35 \times 10^{-11}$; head-group Spearman $p = 9.57 \times 10^{-51}$	heads as units, as descriptive conditional permutation summaries (heads share target heads, prompts and the model): exact $p = 0.0143$; $\rho = -1$, $p = 2.76 \times 10^{-7}$ one-sided, 5.51×10^{-7} two-sided	configurations sharing source heads, Name Movers and prompts treated as independent	No (inferential unit and test changed)
C12	Operator-slice theorem hypothesis	C Zariski-dense in an irreducible $V(P)$	P irreducible and not an associate of $a - a_0$, the recompute branch an infinite subset of $V(P)$; counterexample $P = (b - ma - c)^2$	irreducibility of the zero set was assumed where irreducibility of P is needed	Yes: theorem scope (the stated affine, normalization and transcendental instances are unchanged)
C13	Equal-Hilbert-profile proposition	condition $(m, c) \neq (0, b_0)$; statements for every d	condition equivalent to $m \neq 0$; incomparability needs $d \geq 2$; intersection dimension 0 for $d < 3$	missing degree restrictions	No (range repair)
C14	k-site chain counterexample	bound 44 vs rank 50	per-edge bound 41 vs exact rank 59 (61 with per-arm edges)	naive sum reported as the bound; floating-point rank	No
C15	Concentration constant; eigengap sentence	intrinsic-dimension Bernstein with $L = \log(2d_{\mathrm{int}}/\delta)$; "ill-conditioned means the relative eigengap is small"	the two-sided bound needs a factor $8\,\mathrm{intdim}$ and holds above a threshold; ill-conditioning does not imply a small eigengap	wrong constant; a false equivalence	No
C16	Appendix edge statistics	82 captures; Spearman -0.936; strong-edge median 130	185 captures; -0.869; 38.7; every edge statistic recomputed	legacy pipeline (unmatched captures, all-row fits) and a changed clamped-baseline definition	Procedure and baseline corrected; qualitative association retained
C17	Unmatched-table medians (Table 4)	for example GPT-2 IOI 7115	6489; every median recomputed with training-only fits	same held-out leak as C01	No (procedure corrected)
C18	Figure labels; the ratio in Figure 1(b)	plotted low end labelled "theory-named"; the text quoted a median range ratio for the plotted capture	plots show held-out order statistics of optimised directions; the panel ratio 1.64 is one capture, the capture median 1.86	plotted quantity and aggregation misdescribed (round-1 replacement label was itself wrong)	Yes: labels and supported direction corrected (descriptive plots retained)

id	item	accepted version	this version	cause	conclusion changed?
C19	Support-matching caption	only the symmetric protocol retains a significant reverse direction	the interior protocol also does (GPT-2 induction $q = 7.68 \times 10^{-50}$; Qwen3.5 induction $q = 5.15 \times 10^{-7}$); q-values nominal	caption contradicted its own table	No
C20	Preregistration status	four pre-registered hypotheses failed	four planned hypotheses did not survive (two failed their criteria, two were withdrawn); dated records exist for two	preregistration status not verifiable from the version history	No
C21	Readout screen	no readout values used; screened trials reported rather than dropped	the analysed sample is readout-conditioned by the screen; the IOI collector did not record dropped configurations	readout-conditioned sample selection; rejected configurations not recorded	Yes: interpretation (sample selection disclosed)
C22	Induction head group	query-mediated in the same sense as IOI	reads through keys (K-composition), which a final-position query chart does not test; relabelled	mechanism mislabelled	Yes: interpretation
C23	Mediator-interaction appendix	normalised interaction 3.38×10^{-4}–0.0366 over 9 configurations	no generating script; removed	unreproducible from released code	Yes: claim withdrawn
C24	Summary-block provenance	information ratio 2.68×10^4 and se ratio 72.3 from a block written by unshipped code	the same values (2.68×10^4, 72.3) regenerated from rows by released code	summary block written by unshipped code	No
C25	One-at-a-time baseline	no systematic baseline in the prediction table; the appendix reported g_C only among strong-edge captures	g_C above $10\times$ in $144/185$; both $143/185$, only g_R $28/185$, only g_C $1/185$, neither $13/185$; both generators scored on the same recompute single-site and joint states	missing systematic baseline	Addition
C26	Scope and model names	"three models and five behaviours"; "Qwen3.5"	ten model $\times$ task cells; GPT-2 small, Qwen3.5-9B-Base, Qwen3-8B	scope statement compressed; model names abbreviated	No (clarification)
C27	"Predicted prospectively"	a prospective prediction	named from training states without readouts; a re-analysis of inspected captures	temporal claim not supported by the record	Yes: interpretation
C28	Bibliography	Lin and Liu at ICML; Makelov et al. with three authors; unstated versions	arXiv preprint; the conference version with four authors; versions stated	one wrong venue, one author list from a different version, two version choices	No
C29	"The misattribution is the blind generator"	$c(a - a_0)(b - b_0)$ is the blind generator of the framework	that is g_C, equal to g_R only modulo additive terms; the link goes through the transport identity	two generators conflated	Yes: interpretation
C30	Path-patching appendix	a path-patching family summary	no generating script; removed	unreproducible from released code	Yes: claim withdrawn
C31	"Fisher information"	partial Fisher information ratio 2.68×10^4	a residualised design-Gram ratio: 2.68×10^4 (median), 2.42×10^4 (ratio of medians); a plug-in homoskedastic Fisher ratio of 1.49×10^4 uses residual variances that are not a validated noise model	a design Gram was called Fisher information	Yes: information interpretation (a fixed-chart geometric gap remains)
C32	Same-family null p-values	binomial p against the nominal escape rate calibrates the procedure	the null runs at a different sample size and captures share prompts; p-values nominal	null built at a different sample size; independence assumed	Yes: interpretation (p-values nominal)
C33	bfloat16 captures	affects the readout only	recorded states also carry bfloat16 rounding (GPT-2 IOI and Qwen3.5 unmatched captures)	model dtype propagates into the recorded states	No (description corrected)
C34	Coefficient-space angles	direction agreement 0.814° and 1.87°	parameterisation-dependent; removed	parameterisation-dependent quantity used against the paper's own rule	Yes: claim withdrawn (auxiliary)
C35	Matched joint family	a product grid	a sparse design on the same α grid	design described by its idealisation	Yes: interpretation and scope (same alpha grid and descriptive counts retained; exact support matching not established)
C36	Cross-seed chart control	"a different prompt and a disjoint donor set" ($103/170$ to $26/120$)	the same prompt in $10/120$ captures; donors drawn with replacement from a shared pool, so content disjointness is not established	a seed change was described as disjoint content	Yes: interpretation (control scope)

id	item	accepted version	this version	cause	conclusion changed?
C37	Gram kernel and blind space	$\ker G_Q = N(Q, \mathcal{F})$ when $R_Q = \operatorname{supp} \mu_Q$	equality holds if the features are continuous with finite second moments (a sufficient condition); in general the Gram kernel is the almost-everywhere blind space	a regularity assumption was left implicit	No (assumption stated)
C38	Davis–Kahan eigengap	relative eigengap 0.127 and 0.0813 at the low-energy boundary; bounds 39.9 and 46.6	those gaps were taken at the top of the spectrum; at the bottom-3 boundary the joint-family gaps are 0.00295 and 0.00317; the bounds are vacuous a fortiori and their numbers withdrawn	eigengap indexed at the top of an ascending spectrum while the bottom eigenspace was compared	No (the bounds were already reported as vacuous; the numbers are withdrawn)
C39	Legacy robustness checks	"the frozen procedure under four charts"; a Taylor row that reproduces the polynomial row "as the proposition requires"	Euclidean ridge with all-row whitening, so not held-out validation; the difference chart is the pooled intervention mean minus clean; the Taylor row duplicates the polynomial design (zero centre) and is removed; RBF centres are a fixed grid	construction and protocol of the legacy checks misdescribed	Yes: claim withdrawn (auxiliary Taylor check); the other rows are relabelled as legacy checks
C40	Failed hypotheses (3) and (4)	(3) "a frozen circuit transfers without degrading", failed because "there is nothing for the transfer to degrade"; (4) a "tuned lens"	(3) the registered hypothesis was reciprocal degradation, and its gate failed; overlapping components do not show that transfer cannot degrade. (4) a linear surrogate of the attention-output unembedding (C57); $113/224$ of its calibration rows fall in the scored half	hypothesis stated in reverse; representation mislabelled	Yes: interpretation (both remain failed hypotheses)
C41	Synthetic positive control	planting a generator "recovers it" with residual 6.92×10^{-7} (worst 0.0185; random 1st percentile 0.439)	the same values measure subspace membership in $\operatorname{span}\{g, ag, bg\}$ in a fixed basis, not recovery of a unique generator	quantity misdescribed	No (description corrected)
C42	Chart-selection experiment	mutual non-domination $103/170$ to $26/120$; the strongest cell $19/35$ to $1/28$; two cells remain significant	recompute versus clamped evidence, not single-site versus joint; different cohorts (a spread-ratio screen drops 15 own-chart captures; the archived cross-seed result has eight cells); $3/8$ cells remain nominally significant; GPT-2 induction was not the strongest cell	comparison, cohorts and a count misdescribed	No (the withdrawal stands; description corrected)
C43	Survivors and the equal-profile proposition	survivors have strong edges, "exactly the non-degeneracy hypothesis" of the proposition	the proposition concerns recompute versus clamped designs on an exact slice; the association is descriptive (Spearman 0.284; $1/9$ survivors below the strong-edge threshold)	an exact-slice result applied by analogy to a different comparison	Yes: interpretation
C44	Edge-strength statistics	separation as a function of "the slope"; "the two designs are nearly the same experiment"	the correlation is with the edge-strength ratio (Spearman -0.896; with the slope -0.949); 107 strong-edge captures; cosines are basis-dependent, so the "same experiment" statement is withdrawn; 2.29 is a median of within-capture ratios over the strong-edge captures	statistic and population misdescribed; a parameterisation-dependent cosine used as evidence	Yes: claim withdrawn (auxiliary 'same experiment' statement)
C45	Three-site extreme; unmatched table	"the least-visible direction" responds orders of magnitude less; the named direction is "unaffected" by the support gap	the exponent 7.7 is minus the median over captures of the per-capture minimum held-out log energy ratio $\log_{10}(E_S^{\mathrm{te}}/E_J^{\mathrm{te}})$; the unmatched table shows that the qualitative result persists ($79/82$), not that the ratio is unaffected	statistic and an invariance misdescribed	No (description corrected)
C46	Joshi et al. author order	Reizinger before Sridhar	the ICML version lists Sridhar before Reizinger	author list from a different version	No
C47	k-site proposition title	"single-site evidence scales linearly, the class polynomially"	a restriction-rank bound linear in the sum of composed degrees; linear in k only when that sum is, as for affine arms	a title stated more generally than the bound	No (scope of the title corrected; the affine statement is unchanged)
C48	Transcendence corollaries	softmax and gated edges have infinite onset	only with nonzero output weight (and, gated, nonzero gate slope); otherwise the edge is affine	necessary nonzero-parameter conditions omitted	No (conditions stated)
C49	Joint product grid	a joint product grid identifies the degree-3 class at every k	only with at least four distinct values per axis; the numerical joint design was uniform random points	missing cardinality condition; design mislabelled	No (condition stated)

id	item	accepted version	this version	cause	conclusion changed?
C50	Normalisation onset for trained edges	where normalisation dominates, the onset is five	five on the analysed normalisation slice only; not an onset claim for a trained edge	an exact slice result transferred to an approximate condition	Yes: interpretation (scope)
C51	Identification-table statistics	se called the interaction coefficients' "standard error"; flip rate defined as the fraction of random half-splits on which "the fitted coefficient" changes sign; SE-vector reduction, coefficient used for flips and aggregation over captures unstated	SE ratio of Euclidean norms of marginal SE vectors; flip rate of the ab coefficient over random half-splits; capture medians	aggregations not stated	No (definitions stated)
C52	Reparameterisation check	the displayed covariant pencil changes by 4.53×10^{-7}	4.53×10^{-7} is an unregularised-pencil check (largest cell median); the covariant check gives 6.14×10^{-9}	a check of a different pencil attached to the displayed one; aggregation mislabelled	No (attribution corrected)
C53	Chart-selection denominator	"16 directions in 768 dimensions"; the maximiser "lands in its null space"	768 dimensions for GPT-2 only; numerically negligible projection onto the estimated range, not exact null membership	model-specific width stated generally; numerical result stated as exact	No (description corrected)
C54	Representation-table null column	one same-family null column without a stated degree (0.378 for raw activations); at degree 2 every share "four orders of magnitude below it"	that column is the degree-1 null; the null is rebuilt per degree (degree-2 raw 0.24); against each row's degree-2 null the ratio of the capture-median null to the capture-median share is 4.50×10^{3}–1.70×10^{5}, not a common four orders; the null uses half-size splits, so it is a descriptive reference; the degree ladder remains	null degree not labelled; degree-2 comparison stated as a common factor against the displayed degree-1 null	Magnitude only (the qualitative degree ladder remains)
C55	Representation null: rationale	"they do not separate": a spread of about an order of magnitude against a twofold chart-refit sensitivity	the random-projection/attention-output-unembedding (C57) ratio of capture medians is 31.1 at degree 2 (1.43 at degree 1); the shipped refit check changes the share by a factor 0.966; rationale and no-separation statement withdrawn; the registered criterion was not established (one model-by-task unit; one fixed pair of random projections), which shows neither absence nor separation of representation effects	rationale not supported by the shipped check and outside the registered comparison rule	Yes: claim withdrawn (numerical rationale and no-separation statement; hypothesis (4) remains failed, its registered criterion not established)
C56	Figure 2 (left): residual decay	smooth decay on transcendental slices read as "the signature of polynomial approximation"; reference line labelled "machine precision"	withdrawn as a diagnostic: in the same artifact the unplotted gated-linear slice, and softmax on narrower spans, fall below the 10^{-13} line within the tested degrees; infinite onsets rest on the transcendence lemma; the line is a reference level, not float64 epsilon	numerical illustration over-read as a diagnostic; reference line mislabelled	Yes: claim withdrawn (auxiliary numerical-signature reading; proved onsets unchanged)
C57	Representation table: the "logit lens" row	row and hypothesis (4) described as a logit lens	a standardized attention-output unembedding: last-token attention outputs, custom standardization instead of the model's final LayerNorm; results concern this implemented representation, not the conventional residual-stream logit lens	implementation mislabelled as the conventional logit lens	Yes: interpretation and scope (the representation results concern the implemented attention-output unembedding, not the conventional residual-stream logit lens)

B Proofs and exact numerics

Write $H_Q(d) = \mathrm{rank}(\mathrm{Res}_Q|_{\mathcal{P}_{\le d}})$ for the *Hilbert profile*, the number of independent constraints the evidence imposes. On an affine slice, recompute and clamped patching can constrain equally many alternatives while missing different ones.

Proposition 5 (Equal Hilbert profile, different blind spaces)**.** *On the affine two-site slice with $e(a) = ma + c$ and $b_0 = ma_0 + c$, let N_R and N_C be the degree-d blind spaces of the recompute and clamped single-site families. (i) $H_{Q_R}(d) = H_{Q_C}(d) = 2d + 1$ for every d. (ii) If $m \neq 0$*

and $d \geq 2$, then $N_R \not\subseteq N_C$ and $N_C \not\subseteq N_R$. (iii) If $m \neq 0$, then $N_R \cap N_C = h \cdot \mathcal{P}_{\leq d-3}$ with $h = (a - a_0)(b - ma - c)(b - b_0)$, of dimension $(d-1)(d-2)/2$ for $d \geq 3$ and 0 for $d < 3$.

Counts alone cannot rank these two *single-site* designs on an exact slice; the proposition says nothing about single-site versus joint evidence in a trained model. Given $b_0 = ma_0 + c$, the workshop condition $(m, c) \neq (0, b_0)$ is equivalent to $m \neq 0$.

Proof. (i) Count the constraints. Each reachable set comprises two lines, whose vanishing polynomials are multiples of a degree-2 generator g. Thus $N = g \cdot \mathcal{P}_{\leq d-2}$ has dimension $d(d-1)/2$ and profile $\binom{d+2}{2} - d(d-1)/2 = 2d + 1$.

(ii) Each generator witnesses incomparability. For $d \geq 2$, $g_R \in N_R$ but $g_R \notin N_C$ because $(b - b_0) \nmid (b - ma - c)$ when $m \neq 0$; symmetrically $g_C \in N_C \setminus N_R$. For $d \leq 1$ both spaces are zero.

(iii) Find the shared blind differences. $p \in N_R \cap N_C$ iff $p = g_R u = g_C v$; cancelling $(a - a_0)$ gives $(b - ma - c)u = (b - b_0)v$. The linear forms are non-associate when $m \neq 0$. Unique factorisation in $\mathbb{R}[a, b]$ then gives $(b - b_0) \mid u$ and $p \in \langle h \rangle$; conversely, multiples of h of degree at most d lie in both spaces. Equivalently, $N_R \cap N_C$ is generated by $\operatorname{lcm}(g_R, g_C)$ [Pistone and Wynn, 1996]. □

Theorem 2. Find the vanishing polynomials on each arm, then intersect. $I(A_0) = (a - a_0)$ because A_0 is an infinite subset of the line $a = a_0$. P vanishes on Γ, so $(P) \subseteq I(\Gamma)$. Conversely, if Q vanishes on Γ and $P \nmid Q$, then P and Q are coprime because P is irreducible, so by Bézout's theorem $V(P) \cap V(Q)$ is finite, contradicting $\Gamma \subseteq V(P) \cap V(Q)$ with Γ infinite. Hence $I(\Gamma) = (P)$. Since P is not an associate of $a - a_0$, the two irreducibles are coprime and $I(A_0 \cup \Gamma) = (a - a_0) \cap (P) = ((a - a_0)P)$, whose nonzero elements have degree at least $1 + \deg P$. □

The workshop version assumed only that Γ is Zariski-dense in an irreducible $V(P)$. Over $\mathbb{R}$ this does not give $I(\Gamma) = (P)$: for $P = (b - ma - c)^2$ the zero set is the irreducible line $b = ma + c$, but $I(\Gamma) = (b - ma - c)$, so the onset is 2, not $1 + \deg P = 3$. The affine and nondegenerate normalisation instances satisfy the corrected hypothesis, so their onsets are unchanged. The transcendental instances are covered by the separate lemma below, and their infinite onsets are also unchanged.

Irreducibility of the normalisation quartic. To apply the operator theorem, rule out lower-degree factors. Clearing the radical in $e(a) = \beta + \omega a / \sqrt{\kappa a^2 + \epsilon}$ gives $P(a, b) = (b - \beta)^2(\kappa a^2 + \epsilon) - \omega^2 a^2$, under $\omega \neq 0$, $\kappa > 0$ and $\epsilon > 0$ (for RMSNorm, ϵ is the model dimension times the normalisation constant). As a quadratic in $X = b - \beta$ over $\mathbb{R}[a]$ it is $(\kappa a^2 + \epsilon)X^2 - \omega^2 a^2$, with discriminant $4\omega^2 a^2(\kappa a^2 + \epsilon)$. It factors over $\mathbb{R}[a]$ only if $\kappa a^2 + \epsilon$ is a square, which it is not, since it has two distinct complex roots; a factorisation with a factor free of X would need a common factor of $\kappa a^2 + \epsilon$ and $\omega^2 a^2$, and there is none. Hence P is irreducible of degree 4 and not an associate of $a - a_0$, and the recompute branch is an infinite subset of $V(P)$, so Theorem 2 gives $d^* = 5$.

Transcendence of the analysed slices. The following lemma rules out exact polynomial blind differences on the stated softmax and gated slices.

Lemma 6. *Let $e(a) = \omega\, \sigma(ga)$ with $\sigma(x) = (1 + \exp(-x))^{-1}$ and real $\omega, g \neq 0$. No nonzero $P \in \mathbb{R}[a, b]$ satisfies $P(a, e(a)) \equiv 0$ on an interval with nonempty interior.*

Proof. A polynomial relation for the sigmoid would imply one for the exponential. Suppose $P(a, e(a)) \equiv 0$ with $P \neq 0$. Then e is algebraic over $\mathbb{R}(a)$: it satisfies a polynomial equation with rational-function coefficients. Since $e = \omega/(1 + \exp(-ga))$, the rational expression $E := \exp(-ga) = \omega/e - 1$ is also algebraic over $\mathbb{R}(a)$. Clearing denominators gives $\sum_{j=0}^{n} p_j(a) \exp(-jga) = 0$ with $p_j \in \mathbb{R}[a]$; divide out any power of E to take $p_0 \neq 0$. This expression is real analytic, so vanishing on an interval with nonempty interior implies vanishing identically. Now let $a \to +\infty$ if $g > 0$, or $a \to -\infty$ if $g < 0$. Terms with $j \geq 1$ decay exponentially while p_j grows at most polynomially, forcing $p_0(a) \to 0$, impossible for a nonzero polynomial. □

A two-key softmax edge $e(a) = \omega \exp(a)/(\exp(a) + \exp(\kappa)) = \omega\, \sigma(a - \kappa)$ with $\omega \neq 0$ has the lemma's form after a shift, so $I(\Gamma) = (0)$ and $d^* = \infty$ on that slice. For a gated linear-attention edge $e(a) = (wa)\, \sigma(ga)$ with $w, g \neq 0$, a relation $P(a, e(a)) \equiv 0$ would make $\sigma(ga) = e/(wa)$ algebraic over $\mathbb{R}(a)$ on a subinterval with nonempty interior avoiding $a = 0$, and the lemma applies again. Both statements concern the analysed slice in the chosen coordinates (Remark 4).

Proposition 3. On arm i the swept coordinate t parameterises the arm, and every coordinate is a polynomial in t of degree at most η_i (a constant has degree $0 \le \eta_i$). Restricting a degree-d hypothesis gives a univariate polynomial of degree at most $d\eta_i$, hence at most $d\eta_i + 1$ independent constraints. The k arms agree at the clean configuration, imposing $k-1$ constraints and reducing $\sum_i (d\eta_i + 1)$ to $1 + d\sum_i \eta_i$. Subtracting the restriction rank from the class dimension gives the bound.

The degrees must include all downstream compositions: with edges $A \to B \to C$ of degree p, patching A leaves b of degree p but c of degree p^2. On a chain with $k = 4$ sites, edge degree 2 and $d = 5$, using the edge degree for every arm gives 41, whereas the exact rank (over a finite field, generic coefficients, edges shared across arms) is 59; the composed bound, 76, holds.

Affine arms attain the bound. Translate the clean state to the origin. Under the feed-forward order of the sites, the arm directions form a triangular matrix with unit diagonal, so an invertible linear map sends the arms to the coordinate axes. On the axes, the restrictions of 1 and x_i^r ($1 \le i \le k$, $1 \le r \le d$) are linearly independent, so the rank is exactly $1 + kd$. A Cartesian product with at least $d + 1$ distinct values per axis determines every polynomial of degree at most d in each variable, so for $d = 3$ four values per axis suffice. At degree 3, single-site evidence then leaves unresolved 30.0% of the class at two sites, 50.0% at three, 62.9% at four and 71.4% at five.

Numerical checks of the exact results. The left panel of Figure 2 corroborates the operator-slice results. On each of three sampling spans, singular value decomposition (SVD) fits a candidate polynomial, the right singular vector for the smallest singular value of the column-scaled design. Its relative residual reaches numerical precision at d^* on the affine and RMSNorm slices, while the plotted softmax residual decreases over the tested degrees. Numerics cannot certify that no polynomial relation exists; for the transcendental slices the proof is the lemma above.

C Glossary

term	meaning
chart	a map from the sites' outputs to low-dimensional coordinates $z = (a, b, \dots)$
reachable set R_Q	the chart states an intervention family produces
blind space $N(Q, \mathcal{F})$	hypothesis differences in the class $\mathcal{F}$ that vanish on all of R_Q
resolving preorder $\succeq_{\mathcal{F}}$	$Q_1 \succeq_{\mathcal{F}} Q_2$ when $N(Q_1, \mathcal{F}) \subseteq N(Q_2, \mathcal{F})$: Q_1 resolves at least what Q_2 resolves
onset d^*	the least polynomial degree at which the blind space is nonzero
recompute / clamped generator	on an affine slice, the degree-2 generators of the respective single-site blind spaces: $g_R = (a - a_0)(b - ma - c)$ and $g_C = (a - a_0)(b - b_0)$
energy $E_Q(h)$	mean square of a hypothesis difference over the family's states, $\theta^\top G_Q \theta$
held-out J/S energy ratio $r(h)$	$E_J(h)/E_S(h)$ on test rows, for h defined from training rows only
S-weak	a direction with a large $r(h)$: much less constrained by single-site than joint evidence
edge strength s	$\mathrm{sd}(b \mid \text{arm } A)/\mathrm{sd}(b \mid \text{arm } B)$ strong if $s \ge 0.10$; computed on training single-site rows for the named direction, and on all single-site rows in the support-matching and IOI analyses
same-family envelope	the null band for optimised directions built by splitting the single-site family (§4)
capture, cell, head pair	a capture is one (site pair, seed) run; a cell is a model × task. In §7, a head pair is a (source head, Name Mover) pair, and a capture is that pair with a prompt
transport contrast T_C	how much B's measured effect changes when A moves from its clean output to u: the four-point contrast $Y(u, v) - Y(u, 0) - Y(0, v) + Y(0, 0)$ of §7
B-arm swing	the range of the readout over the single-site B arm; the denominator of every "% of swing"
working-model identification width	at a joint state, the largest $\lvert T_C \rvert$ inside a fit's nominal 95% Wald region under the working model; summarised by a median over joint states within each capture, then a median over captures; neither an observed error nor a calibrated bound
design gap	the ratio, joint over single-site, of the smallest eigenvalue of the per-observation residualised design Gram of the cross block

D Protocol

Protocol at a glance. The following protocol underlies Sections 4 to 6; support and chart variants are specified with each comparison.

- **Units.** A *capture* is one (site pair, seed) run, the seed selecting prompt and donors; a *cell* is a model $\times$ task. Table 5 has 185 float32 captures in 10 cells and 38 (cell, prompt) groups. Captures share models, tasks and often prompts, so counts are consistency checks, and we make no population claim.
- **Sites and chart.** Two attention-block outputs, each projected on the unembedding difference of the task's two answer tokens, then whitened with statistics from training rows (the archived optimised profiles use all-row statistics, which is equivalent in exact arithmetic; see below). The head-level IOI analysis of §7 uses a different chart.
- **Single-site family S.** 32 donors $\times$ 7 interpolation coefficients $\alpha \in [-0.5, 1.5]$ on each arm: patch A and recompute B, or patch B with A clean. Each row is one intervention, giving 448 rows.
- **Joint family J.** 256 rows on the same α grid, pairing donors and coefficients in a fixed sparse pattern; it is not a full product grid.
- **Class and split.** Polynomials of total degree at most 3 (10 dimensions). Rows are split in half (S: 224/224; J: 128/128); directions are named or fitted on training rows, frozen, and scored on test rows.
- **Screen.** Captures with $\mathrm{sd}(Y_{\mathrm{joint}}) < 0.15$ are excluded. In Sections 4 to 6 this is the only use of readouts, and it makes the analysed sample readout-conditioned.

Families. Each capture patches two sites. A donor–coefficient pair sets a site's output to $o_{\mathrm{clean}} + \alpha\,(o_{\mathrm{donor}} - o_{\mathrm{clean}})$, where o_{donor} is the site's output on the donor prompt. The single-site family has, for each of 32 donors and each of the 7 coefficients $\alpha \in \{-0.5, -\frac{1}{6}, \frac{1}{6}, 0.5, \frac{5}{6}, \frac{7}{6}, 1.5\}$, one row patching A with B recomputed and one row patching B with A clean: 448 rows. The support-matched joint family has 256 rows. Row $r = 16i + j$ ($i, j \in \{0, \ldots, 15\}$) pairs donors $2i$ and $2j$ with the coefficient indices $(i + j) \bmod 7$ and $(3i + j) \bmod 7$; on each axis it uses 112 of the 224 single-site (donor, coefficient) pairs, a sparse subset of their Cartesian product. The query-chart captures of §7 index donors by donor id instead. Both joint families of Appendix E are collected in one pass with the single-site family, on identical prompts, donors and weights.

Training statistics and archived profiles. For the named direction, the edge fit, whitening and column scaling use training rows only. The archived optimised polynomial profiles (Figure 2, right, §6 and Appendices E, F and I) estimate whitening and column scaling on all analysed rows. For a fixed chart and the complete, degree-ordered monomial bases used there, changing to training-row statistics is an invertible lower-triangular change of feature basis with positive diagonal. This leaves the Cholesky-reduced covariant eigenproblem unchanged in exact arithmetic. On identical row partitions, the fitted eigenfunctions therefore agree up to scale, and the held-out profiles and same-family envelopes are unchanged. Numerically this holds only where the held-out energies are resolved: at the clamped-weak end of Appendix F the clamped family has an exact polynomial kernel, so those energies are at rounding level under either choice. This does not cover chart selection itself or the Euclidean-ridge checks of Appendices J and K.

Precision. All support-matched captures behind Table 5 and all IOI query-chart captures are float32. The earlier unmatched captures of GPT-2 on IOI and of Qwen3.5-9B-Base were produced with a bfloat16 model, so their recorded states as well as their readouts carry bfloat16 rounding; the other GPT-2 unmatched captures are float32. Comparisons with those earlier captures therefore also change precision.

Readout screen. A capture is analysed only if $\mathrm{sd}(Y_{\mathrm{joint}}) \geq 0.15$. No readout enters the naming or scoring of a direction, but the analysed sample is conditioned on the readout through this screen. In the IOI query-chart analysis, 36 published and 36 control captures were designed and 31 and 28 kept; the collector dropped captures that failed the screen or raised an exception without recording which.

Same-family envelope. For optimised directions we compare the held-out profile with a same-family envelope. We split S into two stand-in families and run the identical train–freeze–score procedure. Within each cell, we pool the other captures' null profiles and take the 2.5th percentile

of their minima and the 97.5th percentile of their maxima, leaving each capture's own nulls out. In the full-grid single-site–joint comparison, each stand-in family has 112 training and 112 test rows, against $224/224$ for S and $128/128$ for J. These sample sizes differ, and captures share prompts, so the nominal 2.5% escape rate is not established for the tested procedure. We report escape counts descriptively and any p-values as nominal (Appendix A, C32).

Applying the audit to another circuit. (1) Declare $\mathcal{F}$; the chosen polynomial degree limits which hypothesis differences the audit can assess. (2) Sweep the intervention family and record the internal states, not just the readout. (3) Form G_Q and read off its low-energy directions; with a second family, solve the covariant pencil. (4) Before comparing two families, check that their reachable sets span the same region; for optimised directions, compare with a same-family envelope, remembering that its p-values are nominal. (5) For a transport claim, execute the clamped and joint corners and measure the numerical floor by re-executing them in a different batch composition. Only steps (2) and (5) need GPU time.

E Support matching in full

Two further protocols separate the effect of removing extrapolation from that of enlarging the joint family. *Interior* restricts the single-site family to $\alpha \in [0, 1]$ and leaves the joint family at $\alpha = 1$; it needs no new data but is conservative rather than exact. *Symmetric* restricts both families to the interior coefficients, which answers the objection that matching helped only by enlarging the joint family. Across all 185 support-matched captures the reverse end escapes the same-family envelope in $105/185$ captures under the unmatched protocol, $44/185$ under interior, $22/185$ under symmetric and $9/185$ under matched.

Edge strength is associated with a reverse escape under the matched (Spearman 0.284), interior (0.41) and symmetric (0.362) protocols, and only weakly under the unmatched one (0.142). Under the matched protocol the median edge strength is 2.09 for escapes against 0.125 for non-escapes. These correlations, the edge-strength medians and survivor count of §6, and Figure 5 use the edge statistic stored with the matched analysis: the arm-A/arm-B spread ratio computed on all single-site rows rather than on training rows only; the paragraph on the realised edge below uses the training-row statistic s of §5. Figure 4 shows the held-out profiles behind Table 3, and Figure 5 the escape rates and their association with edge strength. Table 4 repeats the g_R columns of Table 5 against the unmatched joint family on the earlier unmatched captures.

The named generator and the one-at-a-time baseline. In the captures, writing $b = ma + c + \varepsilon$ with the recorded clean b_0, the clamped generator equals $m(a - a_0)^2 + (a - a_0)(ma_0 + c - b_0 + \varepsilon)$. Both generators clear $10\times$ in $143/185$ captures, g_R alone in $28/185$, g_C alone in $1/185$, and neither in $13/185$, a net difference of 27. The median $r(g_R)$ is 874 on weak-edge captures and 38.7 on the 107 strong-edge ones, of which $93/107$ exceed $10\times$. Of the surviving reverse escapes of §6, $1/9$ fall below the strong-edge threshold. For the GPT-2 IOI captures, the single-site range of a exceeds its range under the unmatched joint family by a median factor of 1.86, the support gap behind Figure 1.

The realised edge, measured. On all 185 support-matched captures, the absolute Euclidean cosine between the recompute and clamped generators, in the fixed whitened, column-scaled coefficient basis, falls as the training-row edge strength s defined in §5 rises (Spearman -0.896; with the absolute whitened slope it is -0.949). Its median is 0.999 below $s = 0.10$ and 0.968 on the 107 captures at or above it. These cosines are parameterisation-dependent and descriptive; they do not show that the two designs are equivalent. On the strong-edge captures the median held-out ratio is 13.2 for g_C and 38.7 for g_R, and the median within-capture ratio $r(g_R)/r(g_C)$ is 2.29. The design comparison rests on these energy ratios and on Table 5: the one-at-a-time generator accounts for most threshold exceedances in this chart, and the recompute generator has an additional advantage on stronger edges.

model × task	protocol	low end	q	reverse end	q
GPT-2 × IOI	joint at 1	34/34	3.39×10^{-54}	6/34	5.39×10^{-4}
	interior	34/34	3.05×10^{-54}	1/34	1
	symmetric	32/34	2.61×10^{-48}	1/34	1
	matched	34/34	3.39×10^{-54}	2/34	1
GPT-2 × induction	joint at 1	34/35	1.04×10^{-52}	35/35	8.47×10^{-56}
	interior	35/35	8.47×10^{-56}	33/35	7.68×10^{-50}
	symmetric	33/35	7.68×10^{-50}	13/35	1.31×10^{-11}
	matched	33/35	6.91×10^{-50}	3/35	0.509
GPT-2 × capital	joint at 1	19/19	2.18×10^{-30}	15/19	2.63×10^{-20}
	interior	19/19	2.18×10^{-30}	0/19	1
	symmetric	19/19	2.55×10^{-30}	0/19	1
	matched	19/19	2.18×10^{-30}	0/19	1
GPT-2 × greater-than	joint at 1	20/20	7.28×10^{-32}	3/20	0.0259
	interior	20/20	7.28×10^{-32}	0/20	1
	symmetric	20/20	7.28×10^{-32}	1/20	1
	matched	20/20	7.28×10^{-32}	0/20	1
GPT-2 × agreement	joint at 1	8/9	2.69×10^{-12}	5/9	4.52×10^{-6}
	interior	9/9	7.63×10^{-15}	1/9	1
	symmetric	8/9	5.37×10^{-12}	1/9	1
	matched	8/9	2.69×10^{-12}	1/9	1
Qwen3.5-9B × IOI	joint at 1	20/20	7.28×10^{-32}	16/20	9.23×10^{-22}
	interior	20/20	7.28×10^{-32}	1/20	1
	symmetric	18/20	1.58×10^{-26}	3/20	0.117
	matched	20/20	7.28×10^{-32}	3/20	0.13
Qwen3.5-9B × induction	joint at 1	14/16	1.71×10^{-20}	11/16	6.49×10^{-14}
	interior	15/16	7.28×10^{-23}	7/16	5.15×10^{-7}
	symmetric	8/16	1.64×10^{-9}	2/16	0.476
	matched	13/16	3.11×10^{-18}	0/16	1
Qwen3.5-9B × capital	joint at 1	6/6	2.44×10^{-10}	6/6	1.46×10^{-9}
	interior	6/6	2.44×10^{-10}	0/6	1
	symmetric	6/6	4.88×10^{-10}	1/6	0.987
	matched	6/6	2.44×10^{-10}	0/6	1
Qwen3-8B × IOI	joint at 1	14/15	2.73×10^{-21}	7/15	1.65×10^{-7}
	interior	14/15	2.18×10^{-21}	1/15	1
	symmetric	10/15	1.28×10^{-12}	0/15	1
	matched	13/15	7.46×10^{-19}	0/15	1
Qwen3-8B × induction	joint at 1	10/11	3.08×10^{-15}	1/11	0.243
	interior	11/11	7.15×10^{-18}	0/11	1
	symmetric	8/11	7.06×10^{-11}	0/11	1
	matched	10/11	3.08×10^{-15}	0/11	1

Table 3: Escapes of the same-family envelope by the optimised profile's low end and reverse end, by protocol, over all 185 support-matched captures; denominators are the screened captures in each cell. q is a Holm-adjusted binomial p-value against the envelope's nominal 2.5% escape rate, adjusted across cells for each protocol and end; it is nominal (§4). Under the matched protocol no cell retains a nominally significant reverse end; under the interior protocol GPT-2 × induction and Qwen3.5-9B × induction do, and under the symmetric protocol GPT-2 × induction does.

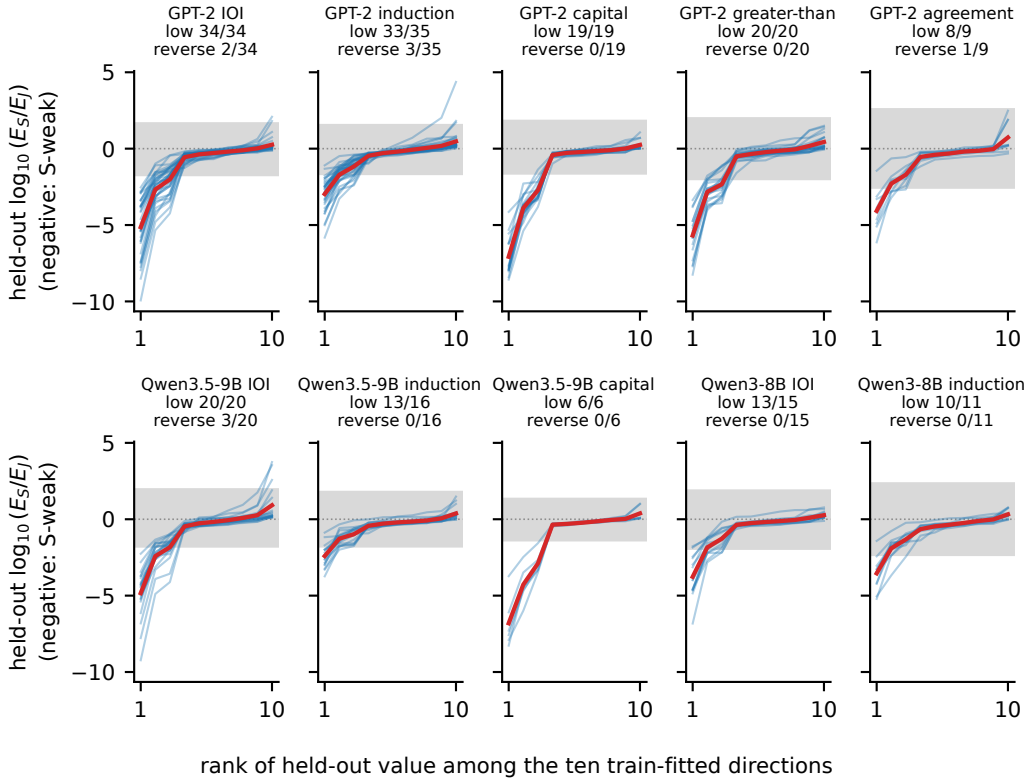


Figure 4: Held-out profiles behind Table 3, support-matched protocol (joint family on the single-site α grid, a sparse design). For each capture (thin blue lines) the ten generalized eigen-directions are fitted on training rows and frozen; the plot shows their held-out $\log_{10}(E_S/E_J)$ energy ratios *sorted by held-out value* (order statistics). Negative values are S-weak: single-site evidence constrains these hypothesis differences less than joint evidence does. The low end is the minimum over the ten frozen directions, not the named generator of §5, and need not be the first training eigenvector. The thick red line is the mean of the held-out order statistics. The grey band applies the percentiles of Appendix D to the same-family nulls of all captures in the cell, pooled, and is shown for display; the escape counts in the panel titles and in Table 3 leave each capture's own nulls out, as in Appendix D. In GPT-2 $\times$ agreement, a reverse endpoint exceeds its leave-one-capture-out upper bound while remaining below the displayed upper bound.

model $\times$ task	median $r(g_R)$	$r(g_R) > 10\times$
GPT-2 $\times$ IOI	6489	12/12
GPT-2 $\times$ induction	101	11/12
GPT-2 $\times$ capital	6413	11/11
GPT-2 $\times$ greater-than	1328	12/12
GPT-2 $\times$ agreement	372	3/3
Qwen3.5-9B $\times$ IOI	371	15/15
Qwen3.5-9B $\times$ induction	63.5	10/12
Qwen3.5-9B $\times$ capital	6768	5/5
all		79/82

Table 4: Held-out J/S energy ratios of the named generator g_R against the unmatched joint family (raw donors, $\alpha = 1$), repeating its columns of Table 5 on the earlier unmatched captures. The qualitative result persists ($79/82$ above $10\times$); the magnitude of the J/S ratio depends on the joint support, and the cohorts differ in precision by cell (Appendix D).

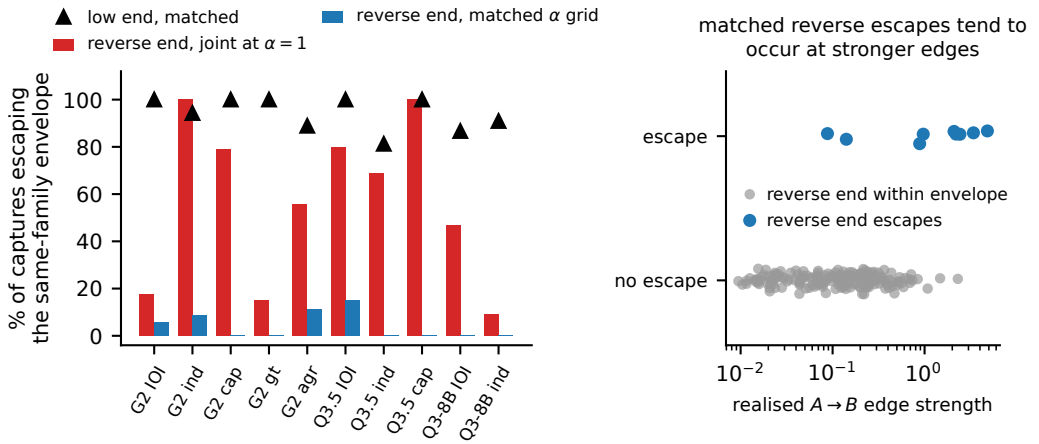


Figure 5: Support matching and escapes of the same-family envelope. *Left:* percentages of captures whose optimised profiles escape at the low (S-weak) or reverse (joint-weak) end: red bars, reverse end with the joint family at $\alpha = 1$; blue bars, reverse end on the matched grid; black triangles, low end on the matched grid. Matching reduces reverse escapes. *Right:* each point is a capture; edge strength (the matched analysis's all-rows statistic, logarithmic axis) is plotted against the binary matched-protocol reverse-escape indicator, with vertical jitter. Blue points escape at the reverse end; grey points do not escape at that end, although their low end may escape. Escapes are associated with stronger edges. Tick labels: G2, GPT-2; Q3.5, Qwen3.5-9B-Base; Q3-8B, Qwen3-8B; ind, induction; cap, capital; gt, greater-than; agr, agreement.

F Chart selection in full

This experiment compares recompute single-site evidence R with clamped single-site evidence C, not single-site with joint evidence. In the unembedding chart the absolute whitened edge slope is small (median 0.0359), so the two designs almost coincide and both ends of their optimised profiles are nominally significant in only $4/10$ cells. We therefore chose the downstream coordinate by maximising a regularised variance ratio between the arms, using states only; the corresponding spread ratio is $\mathrm{sd}(b \mid \text{arm } A)/\mathrm{sd}(b \mid \text{arm } B)$. In that chart the edge is strong (median $|m| = 1.07$). Scoring the held-out R/C energy profile of the train-fitted generalized eigen-directions, the recompute-weak end escapes the same-family envelope in $103/170$ captures, and the clamped-weak end escapes in every retained capture; both ends are nominally significant in $7/10$ cells.

The chart was fitted on states of the capture on which it was scored. In addition to the readout screen, this analysis excludes captures with an arm-A/arm-B spread ratio above 50: 15 captures in the own-chart and donor-holdout analyses. Holding out donors is not enough, because the chart is still fitted on the capture's own prompt; the recompute-weak end still escapes in $75/170$ captures.

With the chart of a different seed of the same layer pair, the recompute-weak end escapes in $26/120$ captures and both ends are nominally significant in $3/8$ cells (GPT-2 capital, GPT-2 IOI and Qwen3.5-9B capital). The cross-seed analysis covers the eight cells with an available source chart, and its artifact predates the Qwen3-8B cross-seed captures, so the cohorts differ and the comparison is not a paired effect estimate. For example, GPT-2 induction falls from $19/35$ to $1/28$.

A seed change does not partition content either: $10/120$ evaluated captures share the chart source's base prompt, and donors are drawn from a shared pool without a cross-seed exclusion rule. This is a cross-seed transfer control, not a disjoint-content validation. The reversal between cells under this control is consistent with selection; the differing cohorts limit its interpretation.

The selection criterion is a variance ratio whose denominator is estimated from at most 16 donor directions in a much larger space (768 dimensions for GPT-2). In the GPT-2 captures for which this was logged, the selected direction had negligible projection onto the span of the donor directions used to estimate the denominator, so its variance ratio was controlled by the regulariser. The chart is therefore not "the direction in which A drives B" but "the direction in which A drives B and these particular donors do not".

G Two planned hypotheses that failed, and two withdrawn claims

Four planned hypotheses did not survive. Two are the withdrawn claims of §6: (1) mutual non-domination of single-site and joint evidence (support confound; Appendix E) and (2) mutual non-domination of recompute and clamped single-site evidence in a chart optimised on the scored states (chart selection; Appendix F). The other two failed their criteria. Hypotheses (3) and (4) have dated preregistration records; for (1), registration and results first enter version control together, so their order cannot be verified; for (2), no dated preregistration entry exists.

(3) Failed hypothesis: a circuit with high in-family faithfulness degrades reciprocally across families. We fitted and froze an edge-attribution circuit under each intervention family and scored it under both. The registered gate required in-family faithfulness of at least 0.85 and a cross-family drop of at least 0.25 in both directions. It failed. The recovered component sets largely overlapped, but overlap alone does not show that transfer cannot degrade. A circuit is a set of components, whereas the ambiguity studied here lives in functional form, which a set cannot express.

(4) Failed hypothesis: interpretability representations have distinctive weakly constrained subspaces. We expected a logit lens or a tuned lens to show S-weak subspaces distinguishable from matched compression controls. The registered success criterion required an effect outside the matched same-family and compression nulls, a nontrivial held-out effect, and a consistent direction across multiple model-by-task units. For 12 GPT-2 captures in a four-dimensional chart, entries below are capture medians of the held-out energy share $v_S = E_S/(E_S + E_J)$ of the least-energy training direction. The same-family null is rebuilt at each degree; each null entry is the capture median of the per-capture 5th percentile over five same-family resamples at the degree indicated by its column header.

The arm we call the attention-output unembedding (the row the accepted version labelled "logit lens"; C57) maps each last-token attention-module output A to $\rho(A) = \big((A - \bar{A})/(\mathrm{sd}(A) + 10^{-6})\big)W_U^\top$, evaluated through its isometric SVD reduction; it uses this custom standardization rather than the model's learned final LayerNorm and acts on attention outputs, not residual-stream states, so it is not the conventional logit lens. The arm labelled "tuned lens" in the accepted version is a least-squares linear surrogate fitted to this attention-output unembedding of the first capture's patch-A-arm rows and reused across captures and sites. Of its calibration rows, $113/224$ fall in that capture's scored half; only the subsequent chart fit is restricted to training rows.

	share v_S		same-family null (5th pct.)	
representation	$\mathcal{F} = \mathcal{P}_{\le 1}$	$\mathcal{F} = \mathcal{P}_{\le 2}$	$\mathcal{F} = \mathcal{P}_{\le 1}$	$\mathcal{F} = \mathcal{P}_{\le 2}$
raw activations	0.143	5.27×10^{-6}	0.378	0.24
attention-output unembedding	0.13	2.01×10^{-6}	0.409	0.342
linear surrogate of it	0.164	2.71×10^{-5}	0.362	0.212
random projection (matched control)	0.185	6.25×10^{-5}	0.379	0.281

At degree 1 each row's capture-median share lies below its degree-1 null by a factor of 2.05–3.15; at degree 2 it lies below its degree-2 null by a factor of 4.50×10^3–1.70×10^5. These factors are ratios of the capture-median null to the capture-median share, not medians of within-capture ratios. Degree-3 values fall below the numerical resolution of the computation, where even an orthogonal reparameterisation moves them substantially, so we do not report them.

The captures cover one model-by-task unit (GPT-2, IOI). The compression control is one fixed pair of random site projections reused across captures rather than a distribution over projections. The captures therefore cannot establish the criterion, which was not met; this shows neither that representation effects are absent nor that any representation separates. At degree 2 the attention-output-unembedding share is below the random-projection share in $12/12$ captures, the raw-activation share in $12/12$ and the surrogate's share in $10/12$; the direction is therefore not specific to the attention-output unembedding, and these are descriptive comparisons within that one unit. The registration treats cross-representation eigenvalue magnitudes as class-relative, so between-row differences in the table do not by themselves establish the registered effect. A sparse-autoencoder arm was omitted by protocol, since no fixed released autoencoder was available for these models. These results concern the implemented representations and do not establish a general conclusion about tuned lenses.

H Per-cell results and stricter checks

Per-cell results. Table 5 gives, for each model × task cell, the median held-out J/S energy ratio $r(g_R)$ of the recompute generator g_R, the number of captures in which that ratio exceeds $10\times$, and the same count for the one-at-a-time (clamped) generator $g_C = (a - a_0)(b - b_0)$. Both generators are scored on the same recompute single-site and joint states, and a large ratio means the direction is S-weak. The $10\times$ threshold is descriptive. GPT-2 is GPT-2 small; Qwen3.5-9B is Qwen3.5-9B-Base.

Donor indices. The row split in §5 places states from the same donor on both sides, since each donor contributes 7 coefficients. Under a common partition of donor indices that scores a joint row only if both of its donor indices are held out, g_R exceeds $10\times$ in $173/185$ captures and g_C in $149/185$ (both $149/185$, only g_R $24/185$, only g_C $0/185$, neither $12/185$). This partition holds out donor indices, not donor content: in $72/185$ captures a held-out donor's patch-A trajectory is identical to a training donor's, and whether the donor prompts coincide was not checked. It is therefore not a disjoint-content validation.

Prompt groups. The 185 captures rest on 38 (cell, prompt) groups that share clean activations, donor pools and task instances, because a capture's seed selects its prompt. Counting a group only if *every* capture in it exceeds $10\times$ under the row split, g_R succeeds in $28/38$ groups and g_C in $17/38$. Resampling the groups with replacement, the 2.5th percentile of the pooled fraction of captures with $r(g_R) > 10$ under the row split is 87.4%. Groups still share models and tasks.

Table 5: Held-out J/S energy ratios of the geometry-named direction g_R, with the one-at-a-time generator g_C as baseline, by model $\times$ task cell. Both generators are scored on the same recompute single-site and joint states. The ratio column gives the median $r(g_R)$; the other columns count captures above $10\times$.

model $\times$ task	median $r(g_R)$	$r(g_R) > 10\times$	$r(g_C) > 10\times$
GPT-2 $\times$ IOI	244	34/34	26/34
GPT-2 $\times$ induction	30.1	30/35	20/35
GPT-2 $\times$ capital	2516	19/19	19/19
GPT-2 $\times$ greater-than	562	20/20	20/20
GPT-2 $\times$ agreement	128	8/9	8/9
Qwen3.5-9B $\times$ IOI	117	20/20	15/20
Qwen3.5-9B $\times$ induction	22.4	12/16	9/16
Qwen3.5-9B $\times$ capital	3972	6/6	6/6
Qwen3-8B $\times$ IOI	60.0	12/15	12/15
Qwen3-8B $\times$ induction	66.4	10/11	9/11
all captures		171/185	144/185

Random directions. For 200 Gaussian directions per capture in the training-scaled coefficient basis, the cell medians of the capture-specific 95th percentiles range from 7.02 to 11.4, and g_R exceeds its own capture's 95th percentile in $180/185$ captures (Appendix A, C02).

I Three sites in a trained model

Design. Proposition 3 counts rank on the population reachable set. Here we extend the trained-model energy comparison to three GPT-2 attention blocks: three single-site arms, each site patched alone across the same coefficient grid with the other two left to the network, against a support-matched joint family over donor triples, in float32, with the same chart, split and envelope. The layer triples are 5–7–9, 5–8–11, 6–9–11 and 7–9–11, with three seeds each.

Full rank, weakly constrained directions. The design matrices have full numerical rank (singular-value tolerance 10^{-9}) under both families, so the comparison concerns ill-conditioning. The profile reports single-site/joint energy, so more negative log ratios mean weaker single-site evidence. For each capture we score every frozen training eigenvector and take the minimum held-out log energy ratio $\log_{10}(E_S^{\mathrm{te}}/E_J^{\mathrm{te}})$. In the 20-dimensional degree-3 class the median over captures of these minima is -7.7; the profile minimum escapes the same-family envelope in $12/12$ captures and the profile maximum in $0/12$. At degree 2 (10 dimensions) the corresponding median is -3.87, and the profile minimum escapes in $12/12$.

J Legacy robustness checks and a synthetic check of the estimator

Charts and classes. These are exploratory legacy checks on the earlier unmatched captures. They use a Euclidean ridge $G_Q + 10^{-10}I$ rather than the covariant pencil, and whitening and feature scaling estimated on all rows; the data-derived charts and the non-polynomial classes therefore also use evaluation states. Only the eigensolve is fitted on the training half, so these are not held-out validations in the sense of §4. Both tables report the median evaluation-half J/S ratio of the least-energy direction.

chart (eight GPT-2 captures)	median J/S ratio, legacy procedure
u (unembedding difference)	3.57×10^6
top activation principal component	7.96×10^7
pooled intervention mean $-$ clean	8.04×10^7
four-dimensional chart (two components per site; degree 2)	7.21×10^5

class	GPT-2 (12 captures)	Qwen3.5-9B (15 captures)
polynomial	1.36×10^7	7.51×10^4
fixed-grid radial basis (RBF; different class)	7.01×10^8	7.91×10^6
random Fourier (different class)	1295	804

The workshop version also reported a Taylor row as an invariance check. It was built with a zero centre, i.e. the identical design matrix, so its exact agreement with the polynomial row is not a check, and it is removed (Appendix A, C39). Magnitudes differ across classes, as Definition 1 predicts, and we never compare them as though the classes shared a geometry.

A synthetic check of subspace membership. In synthetic union-of-slices data with a planted generator g, the pencil estimator's least-energy direction lies close to the known degree-3 blind subspace $\mathrm{span}\{g, ag, bg\}$ in a fixed coefficient basis: its Euclidean projection residual is 6.92×10^{-7} at zero noise and at most 0.0185 up to noise 10^{-1}, against a random-vector 1st percentile of at least 0.439. This checks subspace membership, not recovery of a unique generator, and it is basis-dependent.

K Non-invariant checks and worst-case bounds

Covariant pencil. Under $\varphi \mapsto L\varphi$ every term of the pencil of §4 transforms as $L(\cdot)L^\top$, so its spectrum is invariant in exact arithmetic. In a numerical check of the unregularised pencil on the earlier unmatched captures, the largest cell median of the per-capture maximum relative eigenvalue change under a reparameterisation is 4.53×10^{-7}. A Euclidean ridge on G_Q and Euclidean coefficient angles or cosines are not invariant; they are used only in Appendices E, J and K, as descriptive, legacy or non-invariance checks.

Euclidean ridge. $G_Q + \lambda I$ does not transform covariantly under $\varphi \mapsto L\varphi$ for non-orthogonal L. Under a reparameterisation of condition number 50 on the Qwen3.5-9B IOI captures, the $\lambda = 10^{-6}$ spectrum moves by a median relative 0.867, against 6.14×10^{-9} for the covariant regulariser of §4.

Principal angles. Euclidean angles between coefficient subspaces depend on the parameterisation of the coefficient space. On the earlier unmatched captures, the principal angles between the low-energy eigenspaces of the single-site and joint training Grams have a median of $68.7°$. Recomputing both eigenspaces after a random reparameterisation of the same condition number gives $1.21°$. Both summaries are medians over model–task cells of within-cell medians of per-capture median angles. We report no principal-angle quantity as evidence; the workshop version used such angles as a stability check, and they are removed (Appendix A, C34).

Worst-case spectral bounds are vacuous here. Treating the finite intervention grid as a population, a matrix Bernstein inequality [Tropp, 2015] bounds $\|\widehat{G}_Q - G_Q\|$; for a two-sided deviation of a Hermitian sum the intrinsic-dimension version carries a factor $8\,\mathrm{intdim}$ and holds only above a threshold deviation (C15). The Davis–Kahan bound [Yu et al., 2015] then limits $\|\sin\Theta\|_F$ for the bottom-k eigenspace ($k = 3$) by $2\sqrt{k}\,\|\widehat{G} - G\|/\gamma_k$, where $\gamma_k = \lambda_{k+1} - \lambda_k$ in increasing order. For the joint family the median relative gap at that boundary is 0.00295 (GPT-2) and 0.00317 (Qwen3.5-9B). The workshop version took the gap at the top of the spectrum and used the understated constant, and both corrections enlarge the bound, so it is vacuous a fortiori; its numerical bounds are withdrawn (C38). The measured log–log concentration slopes of the single-site family, -0.3 and -0.486 against a theoretical -0.5, are consistent with the rate but span under one decade of n. Nothing here is used as a guarantee.

Held-out energies are not certificates. For a direction frozen on training rows, $v^\top \widehat{G}_{\mathrm{test}} v$ is an out-of-fit mean over the complementary rows and needs no eigengap. The usual certificate conditions on training rows and applies a fixed-vector concentration inequality to an independent test sample. It is not directly available here: test rows are the deterministic complement of training rows in a fixed finite grid. We make no certification claim.

Table 6: Fit, prediction, Wald envelopes and coverage of the working model on the published IOI head pairs [exploratory; three base prompts; 31 captures]. Compare the prediction errors with predicting zero, and the envelope widths with their coverage. R^2 rows: median over ten donor partitions of the median over captures; "additive terms only" is a separate additive-only refit. Contrast rows use donor partition seed 0; RMS entries are medians over captures. Predicting 0 has an error equal to the RMS of the executed T_C over these held-out joint rows, a subset of the states behind the median RMS in the text. For held-out prediction, fits use only rows whose donors are all training donors; a joint row is scored only when both its donors are held out. "Worse than 0" counts captures whose nominal 95% donor-bootstrap interval for the excess error over predicting zero lies above zero (fits held fixed). A *Wald envelope* is, per capture, the median over the 256 joint states of the largest $|T_C|$ inside the fit's 95% Wald region for γ (point estimate plus radius, all rows of the family); envelope entries are medians over captures, with the percentage taken per capture. It is a working-model identification width that reflects the design, the misspecified fit and its residual variance, not an observed error or a bound on the network. Coverage counts captures in which, at every joint state, the executed $|T_C|$ lies inside that state's envelope.

	single-site fit	joint fit
held-out R^2 on joint readouts, full cubic	-439	0.42
held-out R^2 on joint readouts, additive terms only	0.70	0.60
held-out RMS error predicting T_C, logits (predicting 0: 0.010)	7.45	0.043
fits predicting T_C worse than 0	31/31	22/31
Wald envelope, logits	0.511	0.0129
Wald envelope, % of B-arm swing	31.1	0.764
coverage of the executed $\lvert T_C\rvert$	12/31	1/31

L IOI analysis in detail

Design details. The published pairs combine the S-Inhibition heads 7.3, 7.9, 8.6 and 8.10 with the Name Movers 9.6, 9.9 and 10.0. The control pairs combine the early heads 0.5, 1.2, 2.9 and 3.4, which have no claimed IOI role, with the same Name Movers.

The executed contrast in detail. Across the captures on the three base prompts, the interquartile range of the RMS of T_C is 0.00472–0.0133 logits. Normalising each capture's RMS by its mean absolute per-donor effect at $\alpha \approx 1$ (median 0.322 logits) gives a median percentage of 2.62%. The largest single $|T_C|$ in a capture is a median 17.4% of that effect, and 0.30 logits at most.

The working model in detail. The median edge R^2 of the fitted edge is 0.111, so single-site data inform the cross coefficients only through departures from the affine edge; the low edge R^2 does not by itself show that the scatter is small, and the relevant scale comparison is the ratio 0.0702 of the standard deviation of b on the patch-A arm to that on the patch-B arm. The design gap, whose median in the shared chart is 2.68×10^4, is 5977 if each family is standardised separately. The single-site cubic's held-out R^2 on joint readouts is negative in at least 28 of 31 captures under each of ten donor partitions. For the detection analysis, the ratio of arm-A to arm-B residual variance is a median 0.522 for published heads and 0.00231 for controls, which contradicts the equal-variance assumption of the pooled covariance.

Covariance sensitivity and the joint test. Under a pooled covariance, single-site evidence rejects $\gamma = 0$ in 26/31 published captures against 0/28 controls. Under five alternative covariances controls reject in 8–16 of 28 and published heads in 26–29 of 31. These are covariance-sensitivity calculations for a misspecified model, not a validated calibration. The replication shows the same sensitivity to the covariance estimator (Appendix O). The joint fit, with the marginal-sum donor covariance $V_{\mathrm{up}} + V_{\mathrm{down}}$ (which omits the intersection term and cross-role donor dependence; its calibration is unknown), rejects in 1/31 published and 1/28 control captures.

Estimands of the two tests. The single-site and joint tests of §7 need not target the same coefficients. Correct specification of the cubic model on both supports guarantees a common γ. With the misspecification shown in Table 6, each fit instead approximates the network on its own support: near the two arms for single-site evidence and over the filled region for joint evidence. Sharing a

chart therefore does not guarantee a common estimand. With the pooled working-model covariance $V_S + V_J$, the single-site estimate γ_S is inconsistent with the joint estimate γ_J in $26/26$ single-site rejections ($27/31$ overall), and the joint test's nominal plug-in power at γ_S rounds to 1: had γ_S been the joint fit's target, the joint test would almost always have rejected.

Relation to the workshop version. The Wald-envelope percentages here replace the workshop version's, which evaluated two of the four terms of the contrast and named the wrong denominator (Appendix A, C05–C07).

Working-model identification widths. The widths (computed from nominal Wald regions) of single-site working-model fits vary with degree, covariance and prompt. Percentages are medians over captures (all 31 published captures, or those of one prompt where stated) of each capture's single-site envelope divided by its B-arm swing and expressed as a percentage. By degree: 6.68 (degree 2), 31.1 (3), 220 (4); the single-site envelope exceeds the joint one in every capture at every degree, and the smallest ratio of medians over degrees is 23.6. By covariance at degree 3: pooled 31.1, HC0 (heteroskedasticity-consistent) 27.6, donor-clustered 36.9. By prompt: 146, 15.6 and 21.7; the single-site envelope exceeds the whole swing in $7/12$, $1/10$ and $1/9$ captures of the three prompts.

Each pointwise envelope is the sum of a centre $|D^\top \hat{\gamma}|$ and a Wald radius. Taking medians over joint states and then captures gives centres of 0.169 logits single-site and 0.00232 joint, and radii of 0.337 and 0.0102 logits, respectively. Coverage of the executed $|T_C|$ at all 256 joint states, under $|T_C| \le |D^\top \hat{\gamma}| +$ radius: single-site $11/31$, $12/31$, $15/31$ at degrees 2–4; joint $0/31$, $1/31$, $2/31$. Under $|T_C - D^\top \hat{\gamma}| \le$ radius: single-site $3/31$ and joint $0/31$ at degree 3. The B-arm swing is a median 6.47 times the mean absolute per-donor effect at $\alpha \approx 1$.

Controls. Control head pairs share the Name Movers, prompts and donors with the published pairs. The exploratory captures did not measure the numerical floor, and the replication did not resolve the control family under the registered floor rule (Appendix O), so these data support no published/control transport ratio and no above-control conclusion. Between-prompt and within-prompt standard deviations of $\log$ RMS T_C were 0.438 and 0.682; they informed the replication design.

The edge in the query chart. In this appendix and in §7, the edge strength s of §4 is computed on all single-site rows of a capture, not only on training rows. The median edge strength of the published pairs is $43.3\times$ that of the control pairs (ratio of capture medians). With source heads as units, the four published and four control heads separate completely (exact one-sided Mann–Whitney $p = 0.0143$, the smallest attainable). Each head's value first averages $\log_{10} s$ over prompts whose capture passed the screen, then over its Name Movers. This is a descriptive conditional permutation summary, since the heads share Name Movers, prompts and the model. It is consistent with query mediation and is not a test of it.

Head groups, with heads as units. We repeat the procedure on previous-token $\rightarrow$ induction heads and on the control group. Induction heads read previous-token information through keys; the final-position query chart and final-position patch do not test that mechanism, so this group is a comparison of designs, not a second query-mediated circuit. The median edge strength is 0.0702 for the published IOI pairs, 0.0112 for the induction group and 0.00162 for controls, and the design gap runs the other way: 2.68×10^4, 1.73×10^7 and 1.23×10^{11}. The gap is largest for the controls, whose edge is flattest, so it is a property of the design, not evidence about IOI; the association is largely structural, because a small patch-A spread of b reduces single-site cross-term information. With source heads as units ($n = 10$), the rank correlation between edge and gap is -1 (exact permutation $p = 2.76 \times 10^{-7}$ one-sided, 5.51×10^{-7} two-sided), a descriptive conditional permutation summary, since heads share target heads, prompts and the model; the workshop version's configuration-level p-values treated captures that share heads and prompts as independent. For the induction group, single-site evidence rejects $\gamma = 0$ in $1/18$ captures under a pooled covariance and in 7–10 of 18 under the alternative covariances; the marginal-sum joint test rejects in $0/18$.

Table 7 returns to the unembedding chart for the eight GPT-2 and Qwen3.5-9B cells, reporting design information for the cross coefficients.

model × task	design info., S	design info., J	SE ratio S/J	flip rate S	flip rate J
GPT-2 × IOI	2.95×10^{-4}	0.443	19.3	0.237	0.5
GPT-2 × induction	0.00457	0.647	4.43	0.075	0.25
GPT-2 × capital	2.01×10^{-7}	0.51	618	0.025	0.55
GPT-2 × greater-than	6.69×10^{-6}	0.571	126	0.325	0.3
GPT-2 × agreement	1.20×10^{-4}	0.649	32.4	0.125	0.6
Qwen3.5-9B × IOI	3.17×10^{-4}	0.416	19.9	0.287	0.3
Qwen3.5-9B × induction	0.0116	0.676	4.61	0.35	0.6
Qwen3.5-9B × capital	1.48×10^{-7}	0.526	833	0.0125	0.4

Table 7: Identification of the cross coefficients in the unembedding chart, support-matched captures; entries are medians over captures. *Design info.* is the smallest eigenvalue of the per-observation residualised design Gram of the cross block given the additive block: a property of the design, not Fisher information about the network. The standard-error (SE) ratio is $\|\sqrt{\operatorname{diag} V_S}\|_2 / \|\sqrt{\operatorname{diag} V_J}\|_2$, the ratio of the Euclidean norms of the marginal standard-error vectors in this chart. The flip rate is the fraction of 40 random complementary row-half splits on which the fitted ab coefficient changes sign; it is descriptive and parameterisation-dependent.

M Related work in full

Algebraic experimental design. For a finite design, $I(R_Q)$ is the design ideal and the rank of the evidence map is the Hilbert function of the quotient; aliasing is exactly $N(Q, \mathcal{P}_{\leq d})$. This calculus is due to Pistone and Wynn [1996], developed in Pistone et al. [2001] and surveyed by Maruri-Aguilar and Wynn [2012]. The algebra is inherited from these works. Our contribution is the modelling step: under patch-and-recompute the network, not the analyst, produces the design points, which makes R_Q a measurable property of a trained model. Recent work asks interpretability claims to state their intervention class and identification assumptions [Joshi et al., 2026, Lin and Liu, 2026].

Comparison of experiments, and one-at-a-time designs. Ordering experiments by informativeness is due to Blackwell [1953]. For Gaussian linear experiments with known nonsingular noise covariances and a common mean parameter, Blackwell informativeness is equivalent to Loewner order of the information matrices [Hansen and Torgersen, 1974], and Pukelsheim [2006] develops the design theory of Loewner-incomparable pairs. Our resolving preorder coarsens that order to compare only which directions are resolved within $\mathcal{F}$. Single-site patching under clamping is a one-at-a-time design, whose inability to estimate interactions is textbook material [Czitrom, 1999]; the recompute semantics differ from it only through the realised edge (§5).

Causal mediation. Patch-and-recompute measures a total effect; clamping measures a controlled direct effect. Mediation analysis states the assumptions under which each is identified [Pearl, 2001, Robins and Greenland, 1992, VanderWeele, 2015]. VanderWeele [2014] gives the decomposition that separates mediation from interaction. Vig et al. [2020] brought mediation analysis to language models. Kekić et al. [2025] identify joint mean effects from observational and single-variable interventional distributions under a pairwise-confounded additive structural model and equal action supports across regimes. Our range checks and executed contrasts diagnose related obstacles but do not establish those hypotheses. Vaidyanathan et al. [2026] decompose the patching estimand and expose hidden mediator interactions; that concerns one chosen intervention's effect, not a family's resolving power.

Mechanistic interpretability. We ask what a body of intervention evidence can resolve. Méloux et al. [2025] establish that mechanistic explanations are not unique, and Venkatesh and Kurapath [2026] exhibit large equivalence classes of steering vectors. On intervention semantics, Makelov et al. [2024] show that subspace patching can act through dormant pathways. Geiger et al. [2025] supply a general language for describing interventions, and Eslami [2026] studies separating intervention families for structural causal models. Positive guarantees address other targets: Asiaee [2026] certifies a chosen estimand under a stated intervention distribution, and Dhor and Chen [2026] prove identifiability for a model-intrinsic spectral primitive, asking what property of the model is recoverable.

Complementary evaluations concern intervention success and circuit variability. Bhalla et al. [2024] compare interpretability tools by intervention success; Mueller et al. [2025] benchmark circuit discovery. Bayat Makou et al. [2026] and Venkatesh [2026] study how recovered circuits vary with inputs, granularity and architecture.

Our analysis concerns the intervention families used by automated circuit discovery [Conmy et al., 2023] and alignment search [Geiger et al., 2024]. Experimental design for choosing interventions [Tigas et al., 2022] is the constructive counterpart of our negative statement. Our protocol follows the patching-practice literature [Zhang and Nanda, 2024, Heimersheim and Nanda, 2024, Goldowsky-Dill et al., 2023]. Self-repair after ablation [McGrath et al., 2023] and the gap between localisation and editing [Hase et al., 2023] concern what a successful intervention licenses rather than what a family excludes. The tasks are those of Wang et al. [2023], Olsson et al. [2022], Hanna et al. [2023], Meng et al. [2022] and Finlayson et al. [2021].

N Reproducibility and use of AI tools

A claim ledger supplies the reported numerical results and records each number's source artifacts by hash, unit and aggregation. Verification re-derives the recorded numerical results from the stored artifacts and compares their formatted values and artifact hashes. Appendix A is generated separately from the correction ledger, which records an independent verification check for each entry; associated

review files record the adversarial verdicts. The scope and interpretation of each result are checked separately.

The release accompanying this version, at `https://github.com/Ankush-kadu/activation-patching-reachable-state-geometry`, contains the analysis code, the intervention captures including the replication's, the claim ledger, the preregistration with its frozen hashes, the correction ledger and the manuscript sources. Verifying the claims and rebuilding the paper need only a CPU; re-collecting captures needs a GPU. Two summary artifacts of the workshop version had no generating script (the mediator-interaction comparison and the path-patching controls); the corresponding appendix sections are removed.

Large language models assisted with code, protocol drafting, algebraic checking, adversarial review, writing and literature search. Reported measurements come from executed model interventions and from analyses of stored artifacts, not from numerical values supplied in model responses. Numerical results and algebraic claims were checked with the archived verification procedures. The authors reviewed the content and take responsibility for it.

O Preregistered replication of the IOI transport analysis

Registration. The preregistration, the prompt pool, the collection driver and the analysis script were frozen by hash, with an empty sealed output directory, before any fresh forward pass. The analysis ran as registered, with no change after the first data were seen.

Design. GPT-2 small in float32, with the unchanged collector of §7. The 8 IOI templates of Wang et al. [2023] each have a first clause differing from every template used in this paper's previously inspected captures. We converted them to the name order of our captures and drew three prompts per template from fixed seeds, giving 24 prompts. Each prompt has the 12 published and 12 control head pairs of §7 and the same per-capture design (single-site arms, clamped arm, 256 joint states), 576 captures in all. The eight templates are the complete evaluation set and the unit of inference; we report finite-set criteria over them, not population p-values.

Summaries. For each capture let r_c be the RMS of T_C over the tested joint states and $q_c = 100\, r_c/\mathrm{swing}_c$, where swing_c is that capture's B-arm readout range. For prompt p, M_p and M_p^{rel} are the medians of r_c and q_c over its screened published captures; for template t, V_t and V_t^{rel} are the medians of M_p and M_p^{rel} over its three prompts.

template	T1	T2	T3	T4	T5	T6	T7	T8
V_t (logits)	0.0181	0.0135	0.00673	0.0164	0.0118	0.00904	0.0147	0.0121
V_t^{rel} (%)	0.853	0.598	0.519	0.537	0.661	0.771	0.882	0.589

Validity gates.

(V1) Re-collecting all 24 captures of an original seed had to reproduce the stored readouts to within 10^{-4} logits and the state coordinates to a relative 10^{-5}, with equivalent re-execution under the original batching; all matched exactly.

(V2) The single-site, clamped and joint rows were re-executed in reverse order with batch size 7 instead of 16, keeping donor activations and the clean pass fixed; a capture's floor is the RMS of the resulting change in T_C, and a family counts as resolved if at least 90% of its captures exceed ten times their floor.

(V3) Identity checks: the joint and clamped arms share upstream coordinates, the joint and B arms share downstream coordinates, and pinned coordinates do not move.

(V4) At least half of the published captures and at least five templates remain analysable; 27 captures failed the screen, 0 raised errors, and all eight templates were analysable.

(V5) Captures with a B-arm swing below 0.5 logits are excluded and reported; none was.

Registered criteria and outcomes. $\mathrm{H1}_{\mathrm{abs}}$ passed: $V_t < 0.10$ logits was required in at least 7 of 8 templates and observed in $8/8$ (median 0.0128; range 0.00673–0.0181). $\mathrm{H1}_{\mathrm{rel}}$ passed: $V_t^{\mathrm{rel}} < 1\%$

was required in at least 7 of 8 and observed in $8/8$ (median 0.629%; range 0.519–0.882%). Both margins were fixed after the exploratory analysis and are disclosed as such.

H2 assesses the published/control comparison, grouping screened captures by prompt p and Name Mover n. In a group with both families present and a positive control median, $d_{p,n}$ is the log of the median published r_c minus the log of the median control r_c; d_p is the median of the defined $d_{p,n}$ over the prompt's groups and d_t the median of the defined d_p over the template's prompts. H2 requires $d_t > 0$ in at least 7 of 8 templates, a positive median of d_t, and both families resolved under V2; a template without a defined d_t does not count. Published contrasts were resolved in $281/281$ captures (median RMS 0.012 logits; median floor 2.80×10^{-5}) but control contrasts in only $111/268$ (median RMS 2.33×10^{-4}; median floor 2.79×10^{-5}). Under the registered floor rule H2 is floor-limited: we report no published/control ratio and no above-control finding. The registered decision rule upgrades the section only if every criterion passes, so the transport result stays exploratory with its smallness replicated as registered.

Secondary, descriptive. Across screened published captures, q_c has maximum 6.86% and 90th percentile 1.85%; these summarise the RMS over the joint states, not the largest $|T_C|$ at a single state.

The single-site cubic's held-out prediction error for T_C exceeds that of predicting zero in $281/281$ published captures, by point error. Single-site and joint Wald envelopes cover the executed $|T_C|$ at all joint states in $155/281$ and $15/281$ published captures. Single-site evidence rejects $\gamma = 0$ in $223/281$ published and $0/268$ control captures under a pooled covariance, while controls reject in 69–142 of 268 under the alternative covariances; γ_S is inconsistent with γ_J in $225/281$ published captures. The descriptive sensitivity analyses give the same classifications when excluding the template flagged in the registration for sharing the phrase "went to" with an inspected template (T7), or when dropping the screen. These analyses do not change the registered decisions.